

AI-Assisted Discovery

Lecture 2

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Join our Discord!



<https://discord.gg/9Bw5e3zkjk>

Proof verification,
continued

What future papers might look like

Clear indication of:

- What written by human/AI
- What's verified (by Lean)

Example:

<https://arxiv.org/abs/2512.14575>

Finally, the positivity of all descendant invariants $D(\mathbf{e})$ is well known, following e.g. from the recursive formulas for computing these invariants derived from the Witten–Kontsevich theorem [Wit91, Kon92]. $\square$

Remark 3.3. Theorem 1.2(a) and (b) follow immediately from Theorem 3.1 and Proposition 3.2. The formula (1) for the minimal value is then obtained by the string equation argument given in Section 2.

3.1. Slice sequences. The proof of Theorem 3.1 proceeds by analyzing how D varies along “slices” where only two coordinates change.

Definition 3.4 (Slice sequence - [LEAN](#)). Let $D : E(n, d) \rightarrow \mathbb{Q}_{>0}$ satisfy (S), (LC), (P). Given $\mathbf{e} \in E(n, d)$ and distinct indices $i \neq j$, define the *slice sequence* $S : \{0, 1, \dots, q\} \rightarrow \mathbb{Q}_{>0}$ by

$$S_t = D(\mathbf{e}^{(t)}),$$

where $q = e_i + e_j$ and $\mathbf{e}^{(t)}$ is the composition agreeing with $\mathbf{e}$ at all indices $k \notin \{i, j\}$, with $e_i^{(t)} = t$ and $e_j^{(t)} = q - t$.

The key observation is that S inherits strong structural properties from D .

Lemma 3.5 (Palindromicity - [✓](#), [LEAN](#)). *The slice sequence satisfies $S_t = S_{q-t}$ for all $0 \leq t \leq q$.*

Proof. The transposition $\sigma = (i \ j) \in S_n$ exchanges the i -th and j -th coordinates. Since $\mathbf{e}^{(t)} \circ \sigma = \mathbf{e}^{(q-t)}$, symmetry (S) gives $S_t = D(\mathbf{e}^{(t)}) = D(\mathbf{e}^{(q-t)}) = S_{q-t}$. $\square$

Lemma 3.6 (Log-concavity - [✓](#), [LEAN](#)). *The slice sequence satisfies $S_t^2 \geq S_{t-1} \cdot S_{t+1}$ for all $1 \leq t \leq q-1$.*

Proof. For $1 \leq t \leq q-1$, the composition $\mathbf{e}^{(t)}$ has $e_i^{(t)} = t \geq 1$ and $e_j^{(t)} = q - t \geq 1$. Applying condition (LC) to $\mathbf{e}^{(t)}$ with indices i, j yields

$$D(\mathbf{e}^{(t)})^2 \geq D(\mathbf{e}^{(t)} - \delta_i + \delta_j) \cdot D(\mathbf{e}^{(t)} + \delta_i - \delta_j) = D(\mathbf{e}^{(t-1)}) \cdot D(\mathbf{e}^{(t+1)}),$$

which is exactly $S_t^2 \geq S_{t-1} \cdot S_{t+1}$. $\square$

3.2. Unimodality of log-concave palindromic sequences. The following general lemma is the heart of the argument.

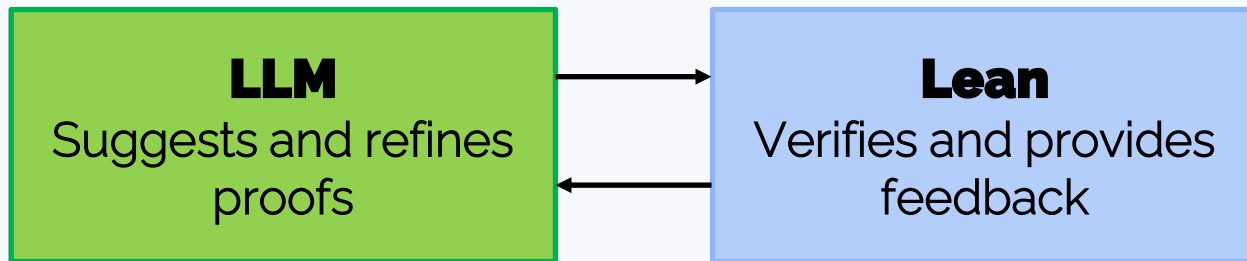
Lemma 3.7 (Unimodality - [✓](#), [LEAN](#)). *Let $S : \{0, 1, \dots, q\} \rightarrow \mathbb{Q}_{>0}$ be a positive sequence satisfying:*

- *Palindromicity:* $S_t = S_{q-t}$ for all $0 \leq t \leq q$.
- *Log-concavity:* $S_t^2 \geq S_{t-1} \cdot S_{t+1}$ for all $1 \leq t \leq q-1$.

Then S is unimodal with maximum at the center:

- (i) $S_t \leq S_{t+1}$ whenever $2t < q$.
- (ii) $S_t \leq S_{t-1}$ whenever $2t > q$.

A dream pipeline



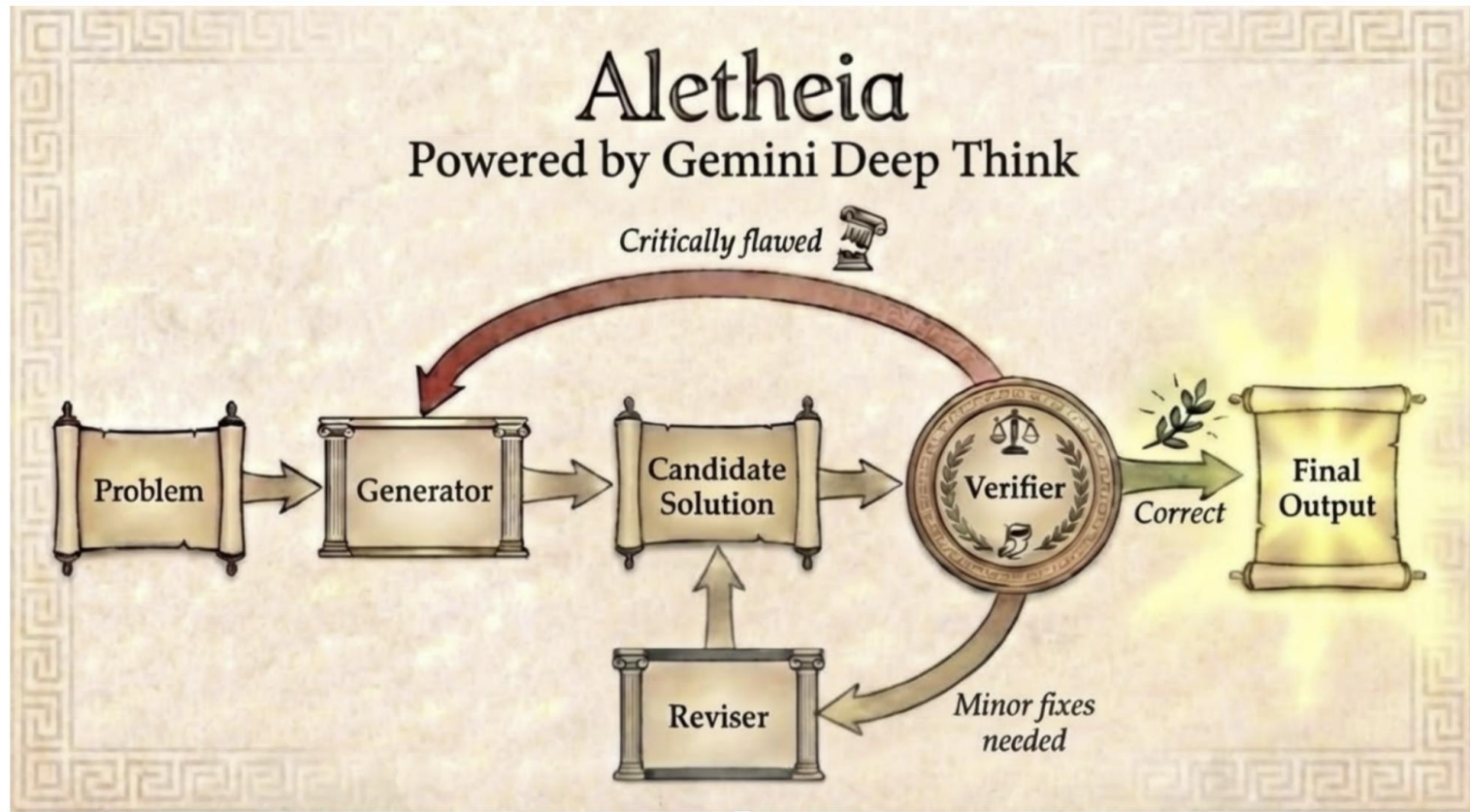


Figure 1 | Overview of *Aletheia*, a math research agent powered by Deep Think that can iteratively generate, verify, and revise for research-level math problems.

Theorem proving in Lean

AxiomProver – multi-agent system that creates proofs directly in Lean.
The recent batch:

- Fel's Conjecture on Syzygies of Numerical Semigroups <https://arxiv.org/abs/2602.03716>
Hilbert series of numerical semigroups (number theory)
- Almost all primes are partially regular <https://arxiv.org/abs/2602.05090>
Primes and Bernoulli numbers
- Parity of k -differentials in genus zero and one <https://arxiv.org/abs/2602.03722>

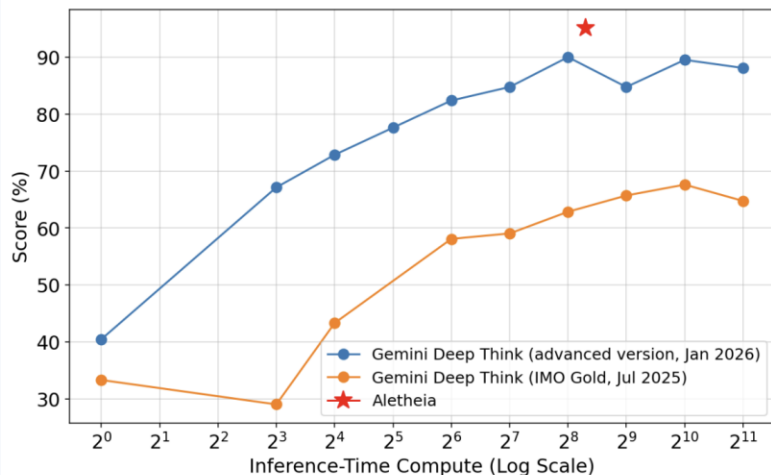
Conjecture 1.1 ([5, Conjecture A.10]). *For odd k and $\gcd(n, k) = \gcd(n + 1, k) = 1$, let $N_k(n)$ be the number of pairs of positive integers (b_1, b_2) such that $b_1, b_2 \leq (k - 1)/2$, $b_1 + b_2 \geq (k + 1)/2$, and $b_2 \equiv nb_1 \pmod{k}$. Then we have*

$$N_k(n) \equiv \left\lfloor \frac{k+1}{4} \right\rfloor \pmod{2}.$$

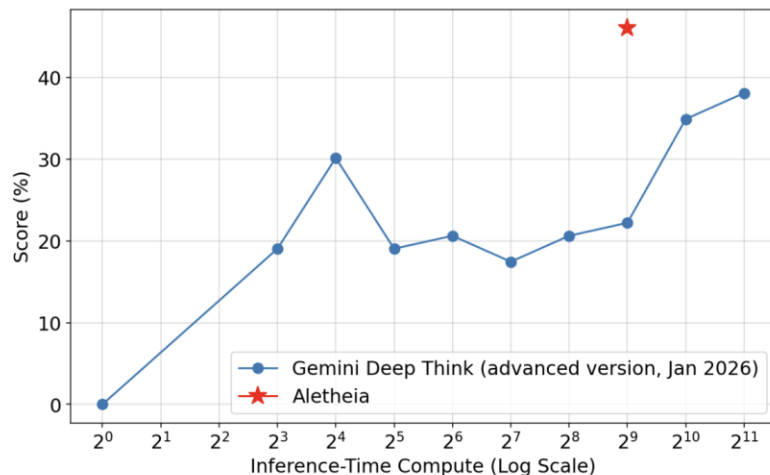
Cons of the dream pipeline

- Even if Lean has something, it might be not the way you want it
- Translation **Latex** $\leftrightarrow$ **Lean** is yet another source of errors. And **Lean** isn't meant for humans
- The dream pipeline is very expensive (proofs in Lean tend to be 2-10x longer than in Latex + you need multiple rounds) and even that, still only copes with elementary problems

Cons of Lean and the dream pipeline: it's expensive



(a) *IMO-ProofBench* Advanced (Olympiad Level).



(b) *FutureMath* Basic (Ph.D. Level Exercises).

Figure 2 | The latest advanced version of Deep Think, as of Jan 2026, has significantly outperformed the IMO-Gold version (Jul 2025) on Olympiad-level problems. The inference-time scaling law also transfers to PhD-level exercises. Aletheia makes further leaps in terms of reasoning quality with lower inference-time compute. All results were graded by human experts.

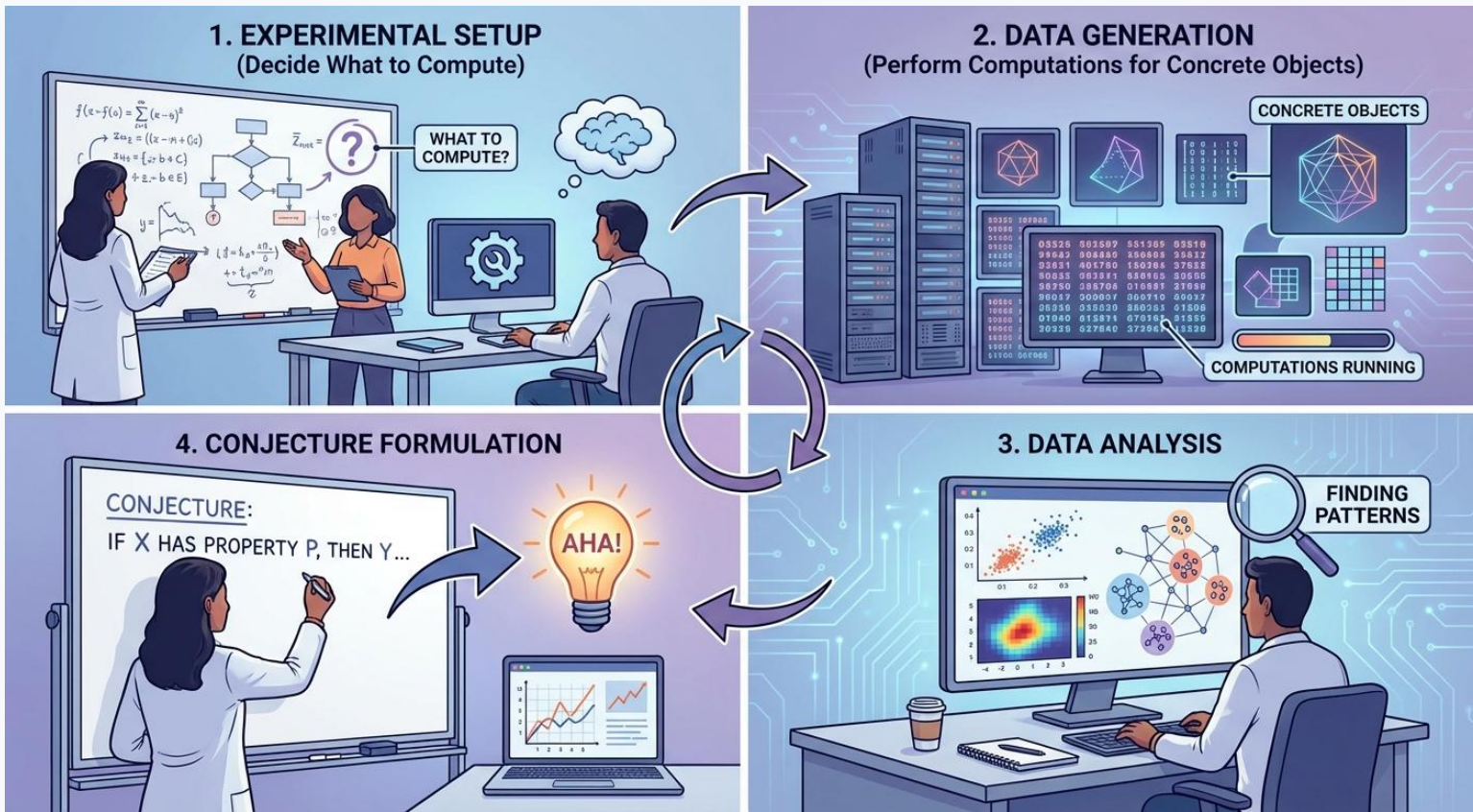
Coding agents time!

Automated discovery pipelines

The discovery loop steps

- Experimental setup
- Data generation
- Data analysis
- Conjecture formulation

The discovery loop steps



distance 0

distance 1

distance 2

distance 3

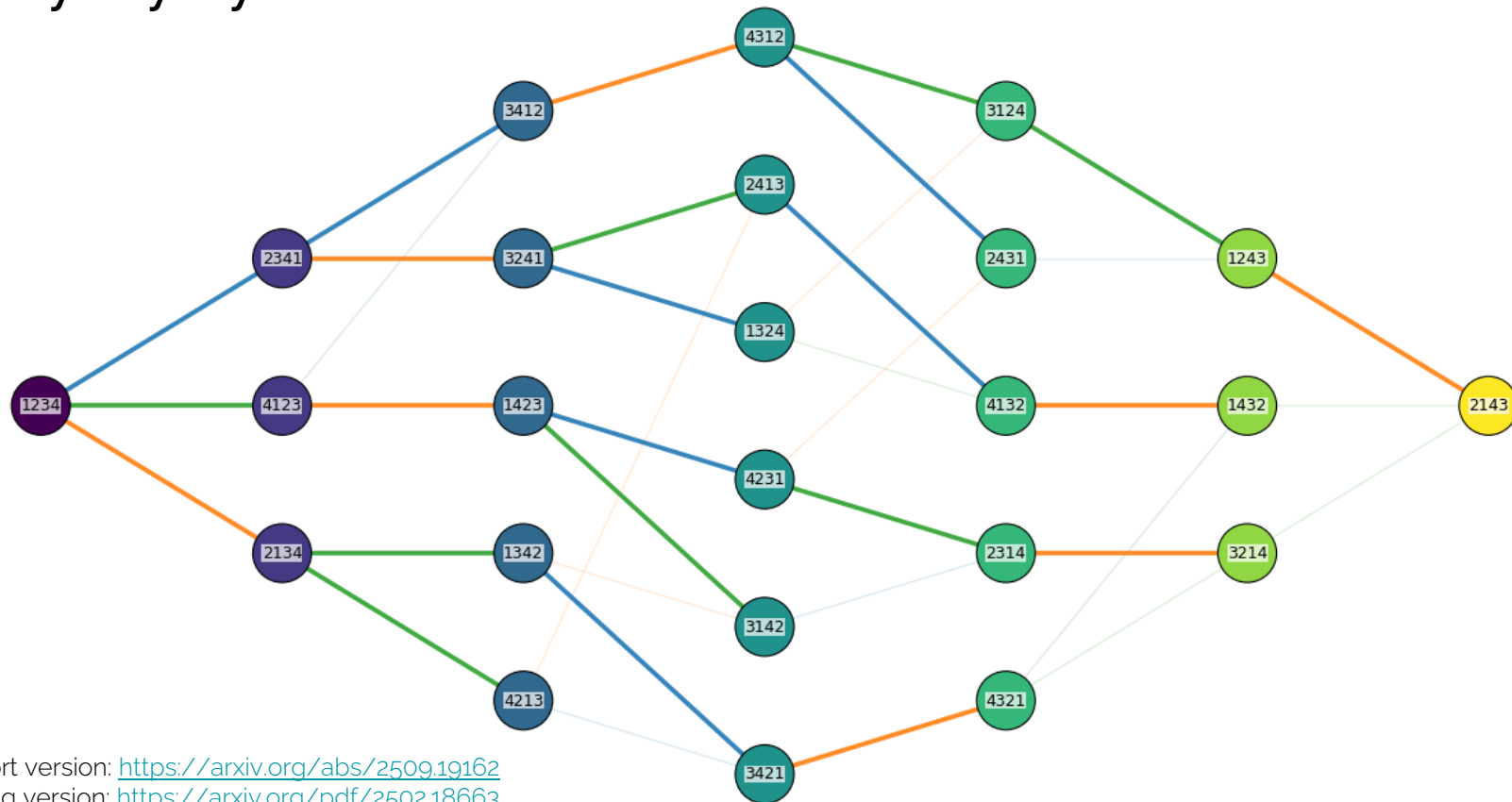
distance 4

distance 5

distance 6

CayleyPy Growth

Layered BFS layout of the Cayley graph of S_4
Generators: (1234), (1432), (12)



Short version: <https://arxiv.org/abs/2509.19162>

Long version: <https://arxiv.org/pdf/2502.18663>

CayleyPy Growth

6. CASE STUDY – CONSECUTIVE 4-CYCLES $(i, i + 1, i + 2, i + 3)$, USE OF AI

Consider generators of S_n given by consecutive 4-cycles $(i, i + 1, i + 2, i + 3)$, $1 \leq i \leq n - 3$ and their inverses. Although these are rather obvious generators, to the best of our knowledge their diameter has not been discussed in the literature. We propose the following conjecture for this set of generators.

- Conjecture 14.** (1) *The diameter is given by $n(n - 1)/6 - 1$, for $n = 3k, 3k + 1, n \geq 6$, and $n(n - 1)/6 + 2/3$, for $n = 3k + 2$, and $n \geq 8$*
- (2) *The longest elements (antipodes) include: $[n, n - 1, \dots, 3, 1, 0, 2]$, for $n = 3k, 3k + 1, n \geq 6$ and $[n, n - 1, \dots, 3, 2, 0, 1]$, for $n = 3k + 2$.*
- (3) *The mean diameter is approximately equal to $0.036n^2$, mode $\approx 0.045n^2$*

CayleyPy Growth

Maximal diameter for Cayley graph of the group S_n (oriented graph)			
n	Maximal diameter	Example of a set of generators	Sheveleva2 params (n,k)
3	3	(01), (02)	
4	7	(01), (123)	
5	14	(01)(23), (0314)	
6	18	(01)(23)(45), (012)(34)	
7	34	(01)(23)(45), (052)(146)	
8	44	(01)(23)(45), (1736)(25)	(8, 2)
9	61	(01)(23)(45), (3647)(12)(58)	(9, 3)
10	83	(01)(23)(45)(67)(89), (185)(237)(469)	
11	93	(01)(23)(45)(67)(89), (1528)(47)(6, 10)	type 2
12	106	(01)(23)(45)(67)(89), (1, 11, 9, 10)(04)(28)(36)	(12, 4)
13	147	(01)(23)(45)(67)(89), (1, 11, 2, 12)(34)(56)(78)(9, 10)	(13, 2)

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distance 0

distance 1

distance 2

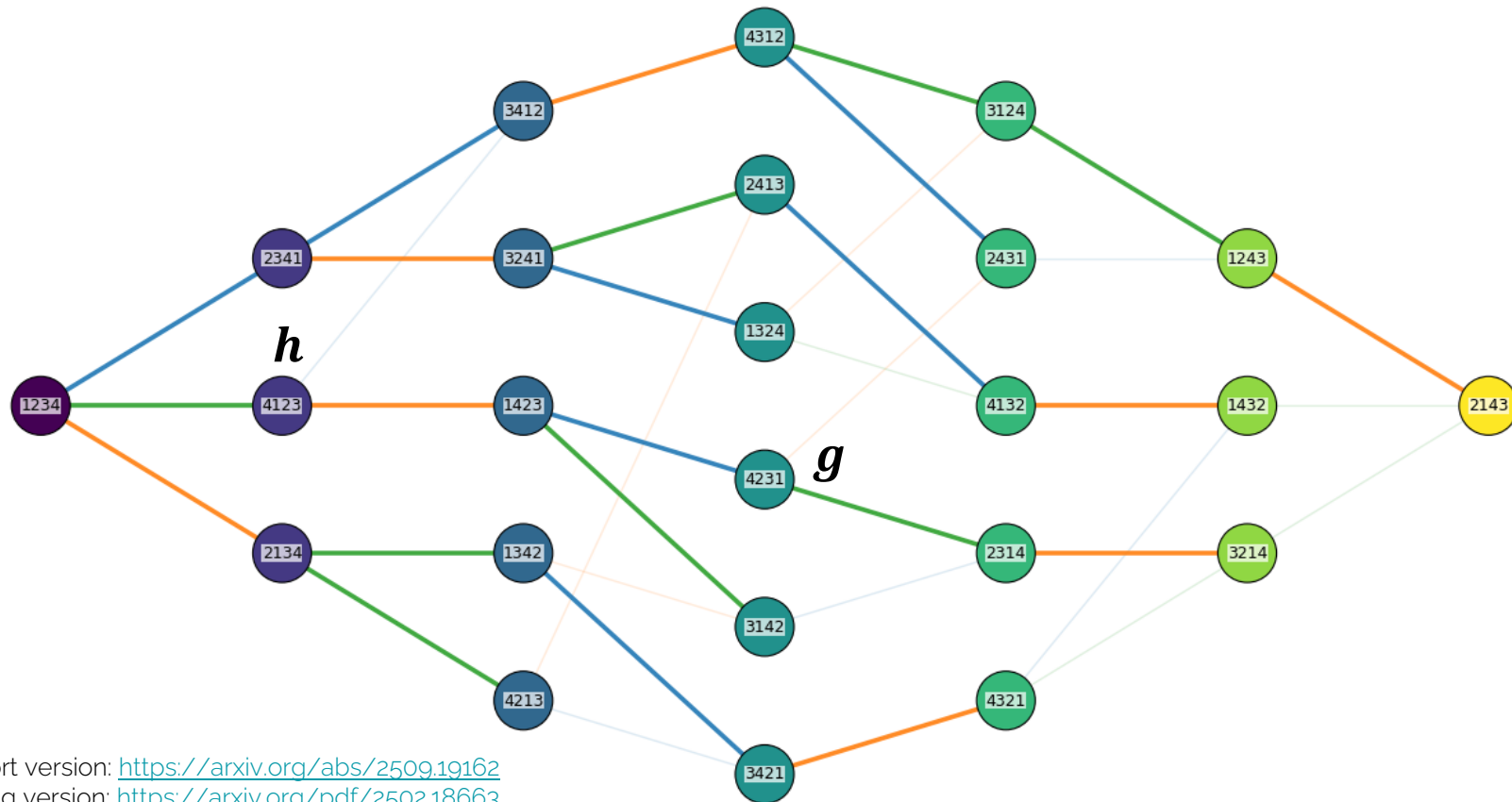
distance 3

distance 4

distance 5

distance 6

Layered BFS layout of the Cayley graph of S_4
Generators: (1234) , (1432) , (12)



distance 0

distance 1

distance 2

distance 3

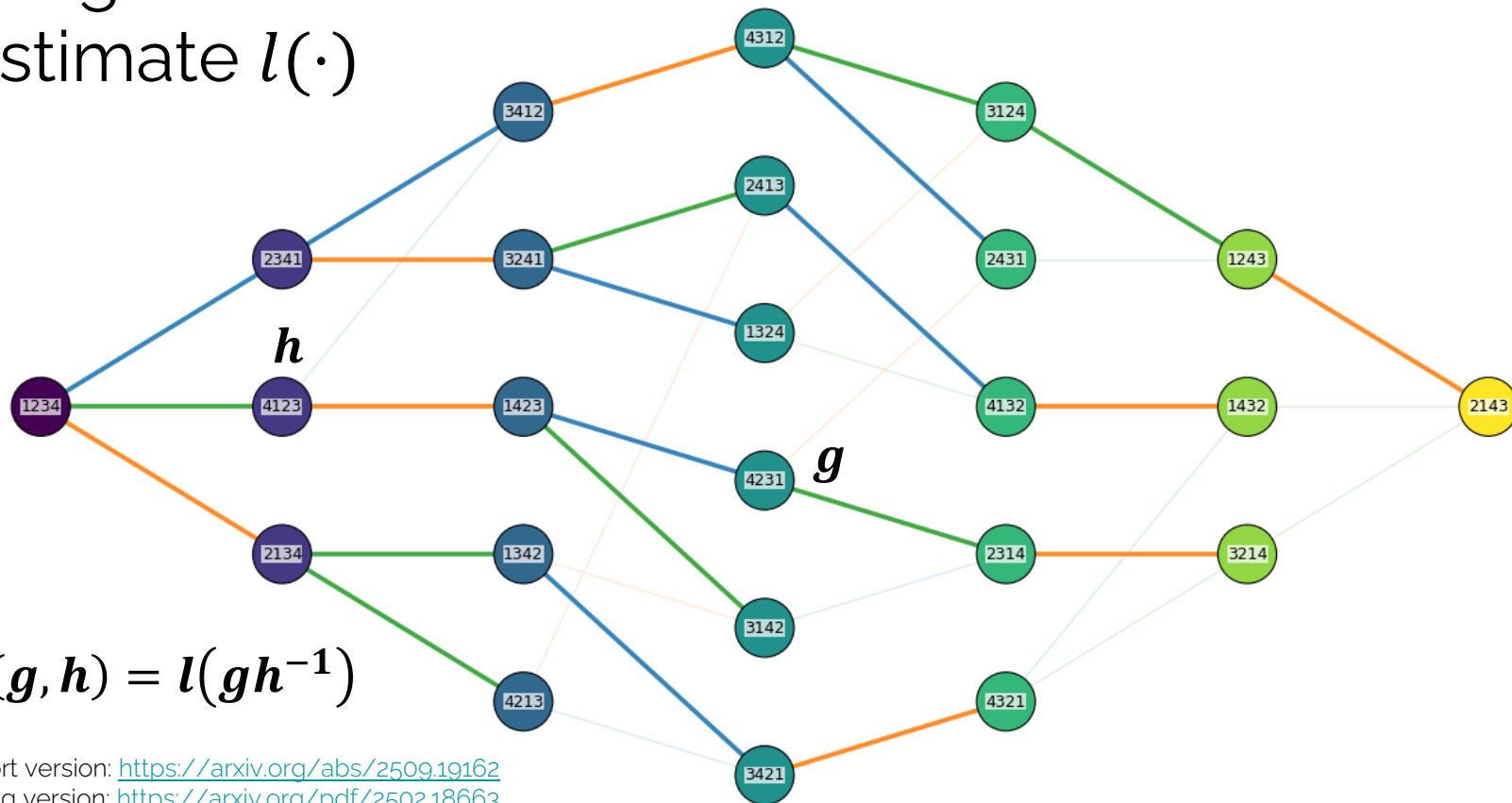
distance 4

distance 5

distance 6

Imagine we can
estimate $l(\cdot)$

Layered BFS layout of the Cayley graph of S_4
Generators: (1234) , (1432) , (12)



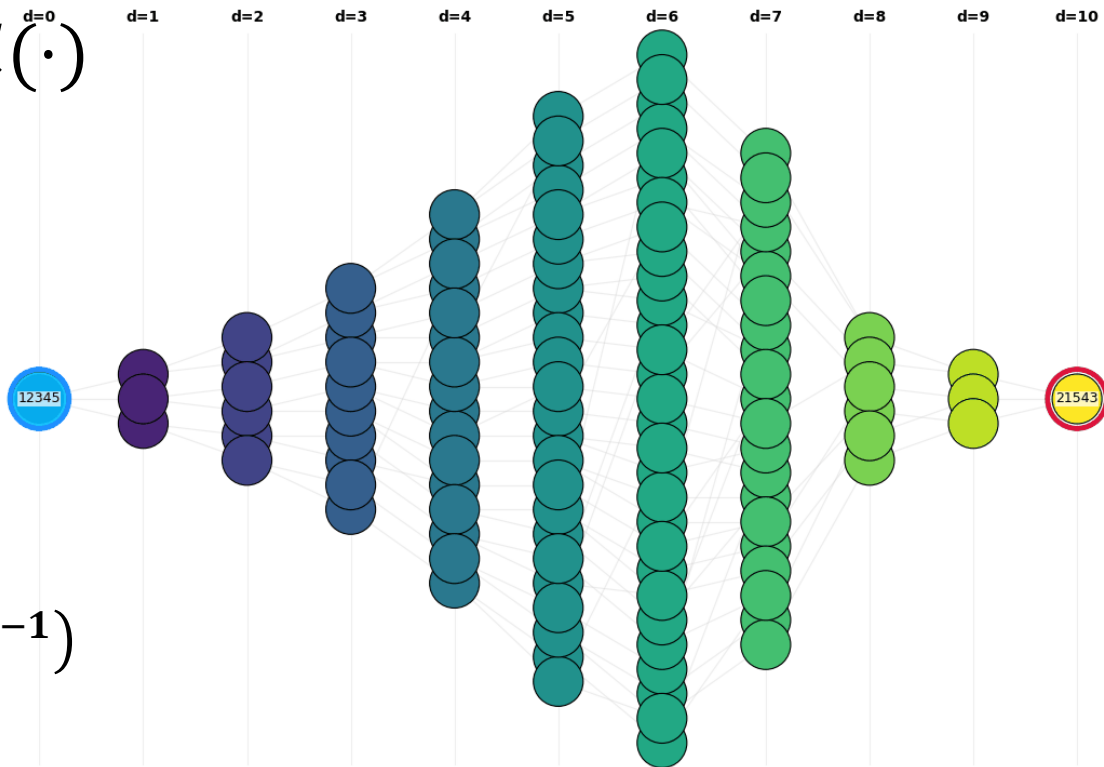
$$d(g, h) = l(gh^{-1})$$

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Imagine we can
estimate $l(\cdot)$

Beam search on the Cayley graph of S_5 (beam width = 2)
Generators: $c=(1\ 2\ 3\ 4\ 5)$, $s=(1\ 2)$ | target=21543 | diameter=10
Phase=start | beam depth=0



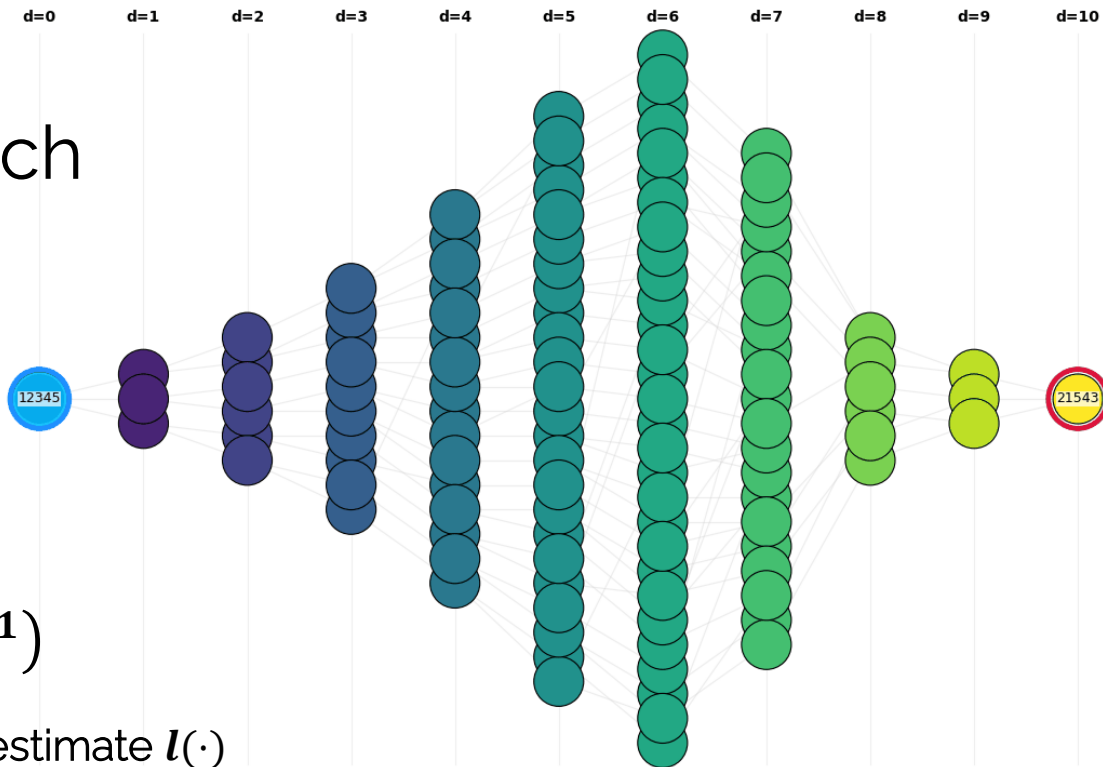
$$d(g, h) = l(gh^{-1})$$

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Beam search on the Cayley graph of S_5 (beam width = 2)
Generators: $c=(1\ 2\ 3\ 4\ 5)$, $s=(1\ 2)$ | target=21543 | diameter=10
Phase=start | beam depth=0

Beam search example



$$d(g, h) = l(gh^{-1})$$

Need a heuristic to estimate $l(\cdot)$

Short version: <https://arxiv.org/abs/2509.19162>

Long version: <https://arxiv.org/pdf/2502.18663>

CayleyPy Growth – the pathfinding algorithm

- 1. Build training data by random walks:** (g, k) , where g is a group element and k is the step number at which g is reached from e .
- 2. Train a predictor:** a neural net $f(g)$ to estimate k .
- 3. Optional RL refinement** to account for the randomness of the initial data
- 4. Guided beam search.** Imagine we need to find the shortest path from e to g . On every step, we keep the **beam** of up to B candidates and:
 - Expand each path $e \rightsquigarrow g_i$ from the beam to B candidates $g_i \rightarrow h_{ij}g_i$
 - Score each h_{ij} by length $l(h_{ij}g_i, g) \approx f(g(h_{ij}g_i)^{-1})$
 - Keep B worthiest successors

Short version: <https://arxiv.org/abs/2509.19162>

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CayleyPy Growth – finding the Cayley graph's diameter

- 1. Perform precise computations for small n .**
- 2. Conjecture the $d(n)$ formula and the antipode based on this data.**
- 3. For large n , find the distance to the conjectured antipode using the smart pathfinding algorithm**

Short version: <https://arxiv.org/abs/2509.19162>

Long version: <https://arxiv.org/pdf/2502.18663>

Hometask

Problem: find the law for the *diameter* of S_n w.r.t. the generators $(1, 2, \dots, n)$ and $(1, n)$.

Use Claude Code or Codex for this:

- Generate data for $n = 3, \dots, 12$ (case $n = 12$ might require **C/C++/Rust**)
- Ask the agent to analyze it and conjecture the law

The answer is:

- For odd $n \geq 5$: $(3n^2 - 8n + 9) / 4$
- For even $n \geq 6$: $(3n^2 - 8n + 12) / 4$

Challenges for an AI agent:

- Clauses
- The formula doesn't hold for $n = 3, 4$

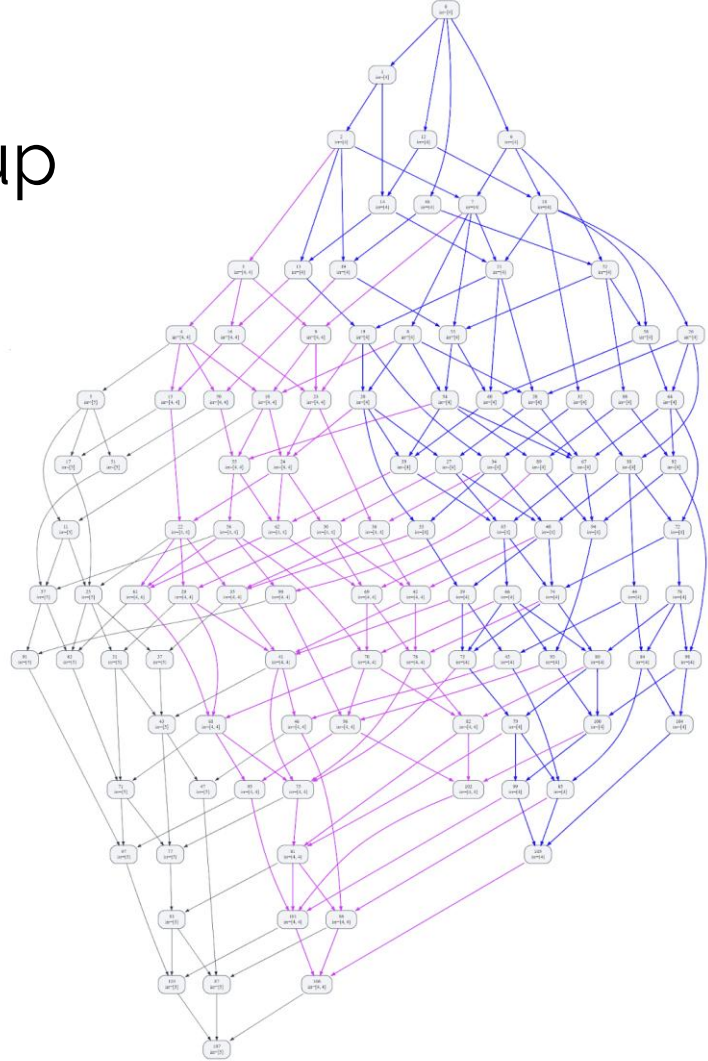
What will be the "aha moment"?

Quiver

Grassmannians

Quiver Grassmannians – the setup

- Sprawling families of objects (multi-projective varieties)
- Very diverse (any projective variety is a quiver Grassmannian)
- So little is known, it's even hard to conjecture anything
- At least for the small one, we can calculate some geometric invariants
- We do “conjecture mining”



Here come coding agents

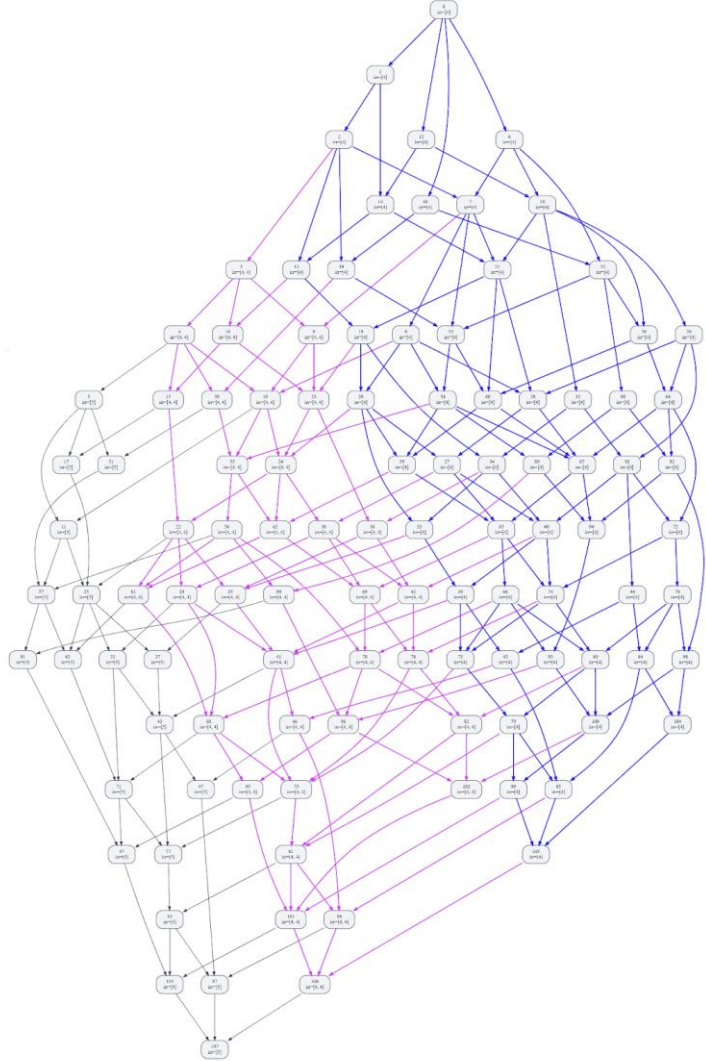
Agents: GPT, Codex, Claude Code

Tools: Python, Macaulay 2, a bit of Sage

Hardware: 128xCPU@Nebius

We've built a library for working with quivers

<https://github.com/st-fedotov/quiver/tree/main>

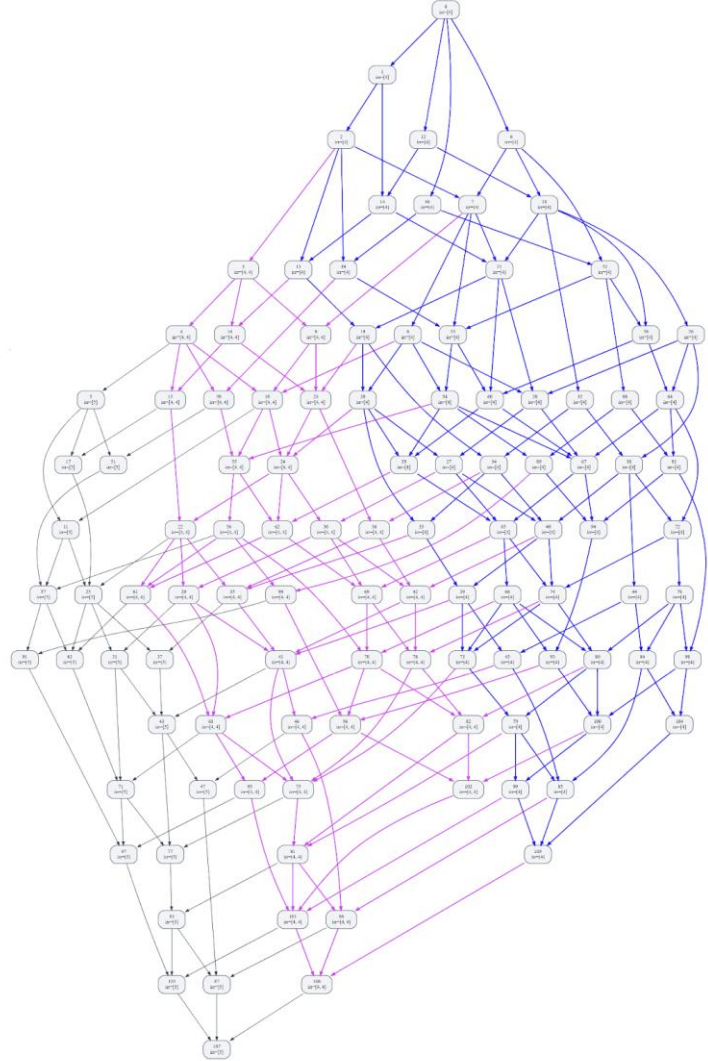


Lessons learnt

1. On the design stage – discussion first, then coding
2. Don't trust your coding agent in dealing with advanced maths
3. When debugging long pipelines, don't let your coding agent fixate on its terminal parts
4. Repetitive helpers bloom like cancer unless you make the agent check what it already has in the library
5. I wish there were **skills** back then

Experimental pipeline

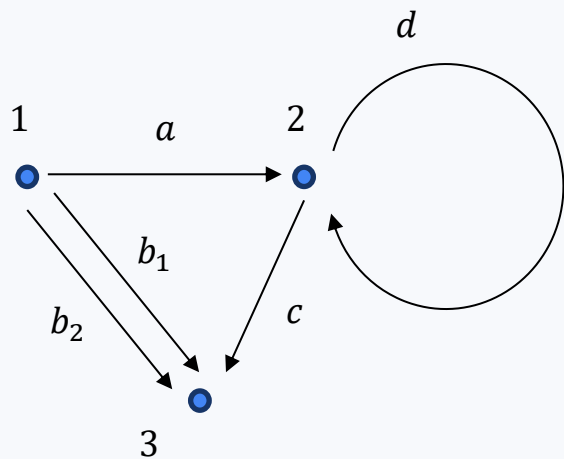
1. Generate data: geometric invariants for particular families of Grassmannians (computationally heavy)
2. Use **Claude Code** to check various things about them, propose and test conjectures
3. Iterate
4. Write a paper 😊



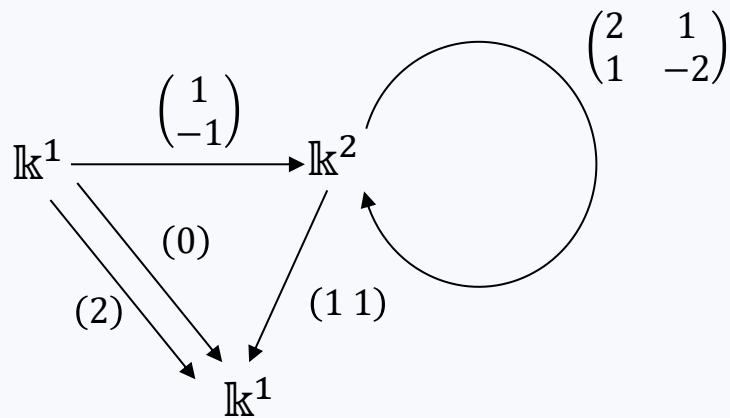
Lessons learnt

1. Claude Code greatly speeds up conjecture search
2. If asked to propose their own conjectures, coding agents to for low hanging fruit (and lack “intuition”)
3. Start with concrete questions; then discuss, ask to propose data augmentation strategies
4. Use skills; don't trust context, document your research process

Quiver



Representation



$$\dim M = (1, 2, 1)$$

Direct sums and indecomposables

$$M: \quad \mathbb{C} \xrightarrow{\begin{pmatrix} 1 \\ -1 \end{pmatrix}} \mathbb{C}^2 \xrightarrow{\begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix}} \mathbb{C}^2$$

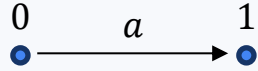
$$N: \quad \mathbb{C} \xrightarrow{(1)} \mathbb{C} \xrightarrow{(3)} \mathbb{C}$$

$$M \oplus N: \quad \mathbb{C}^2 \xrightarrow{\begin{pmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \end{pmatrix}} \mathbb{C}^3 \xrightarrow{\begin{pmatrix} -1 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}} \mathbb{C}^3$$

Indecomposable = no non-trivial direct sum decomposition

Question: can we classify indecomposable representations?

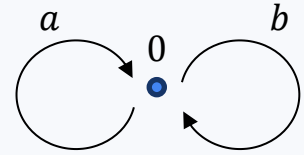
The three cases



Case 1

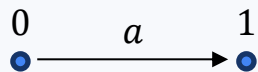


Case 2

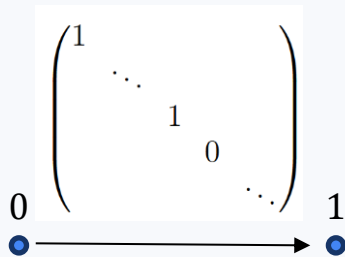


Case 3

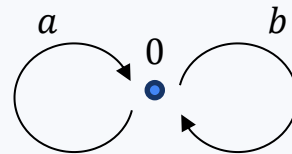
The three cases



Case 1

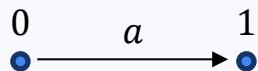


Case 2

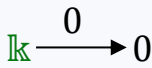
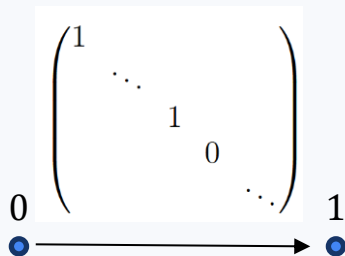


Case 3

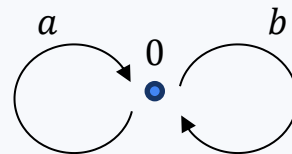
The three cases



Finite

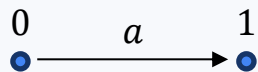


Case 2

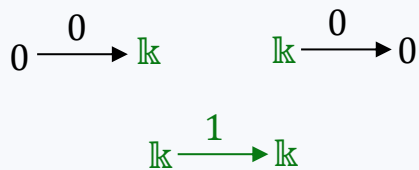
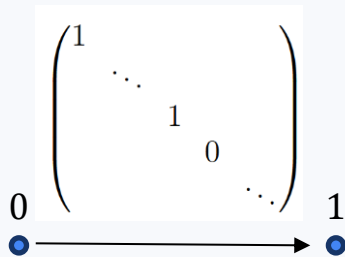


Case 3

The three cases



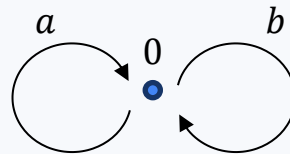
Finite



Tame

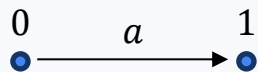
$$\begin{pmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \end{pmatrix}$$

$$J_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}$$

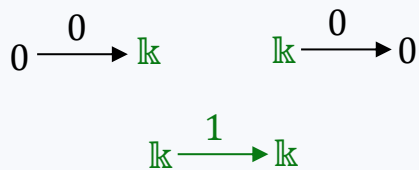
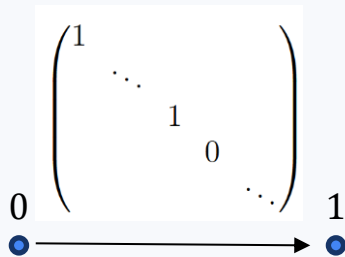


Case 3

The three cases



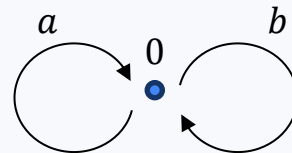
Finite



Tame

$$\begin{pmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \end{pmatrix}$$

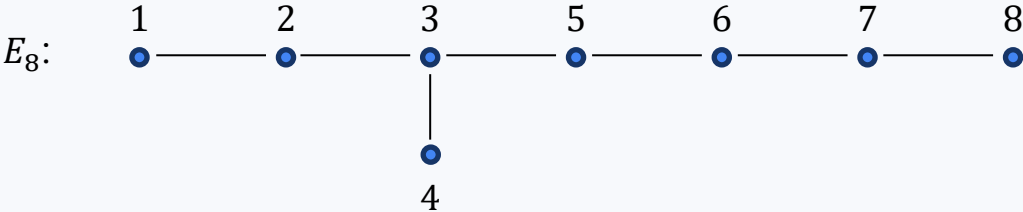
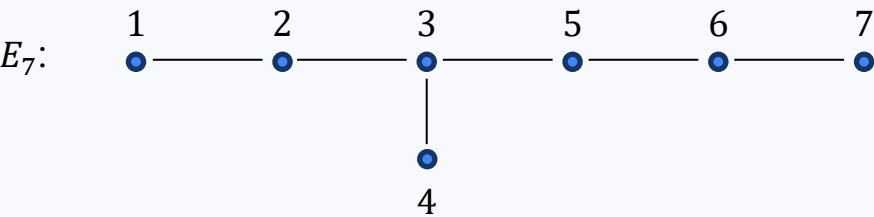
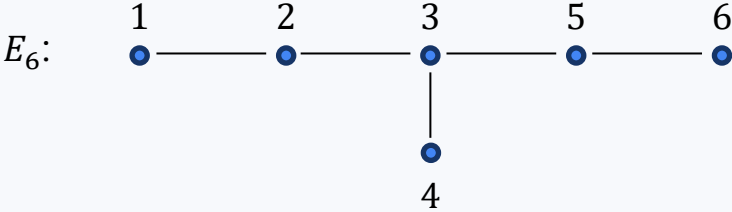
$$J_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}$$



Wild

Finite quivers

Indecomposables =
positive roots



Finite quivers

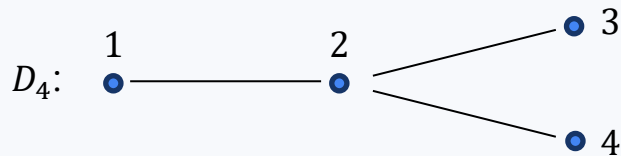


Indecomposables = positive roots

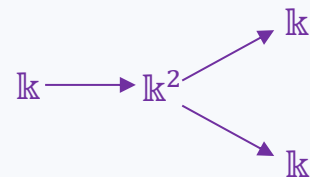
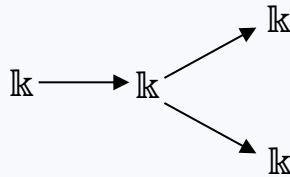
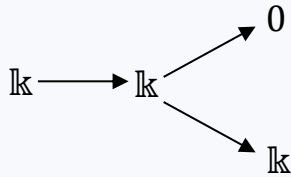
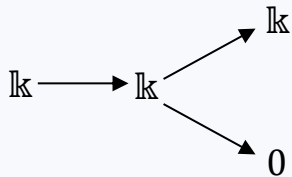
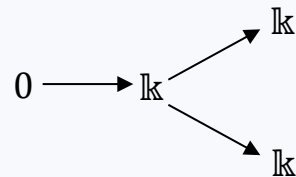
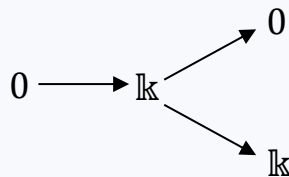
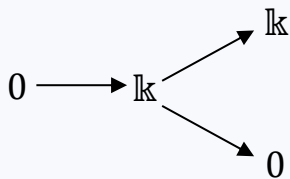
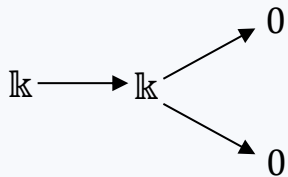
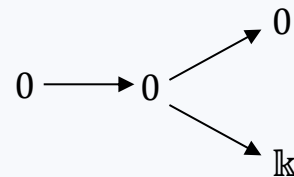
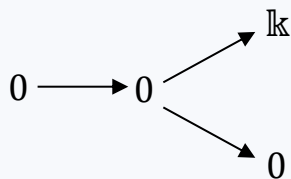
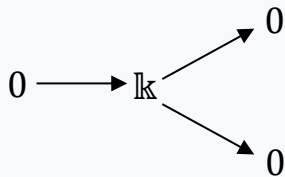
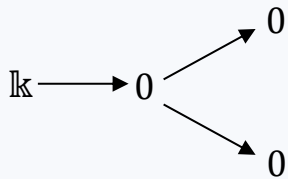
For A_n these are **interval modules**:

$$\dots \xrightarrow{0} 0 \xrightarrow{0} \mathbb{k} \xrightarrow{1} \mathbb{k} \xrightarrow{1} \mathbb{k} \dots \mathbb{k} \xrightarrow{1} \mathbb{k} \xrightarrow{0} 0 \xrightarrow{0} \dots$$

Finite quivers

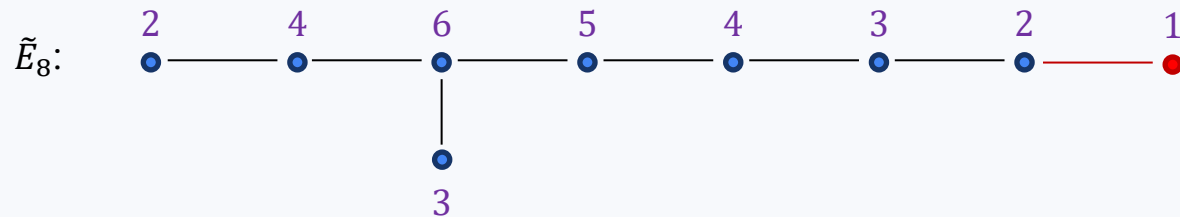
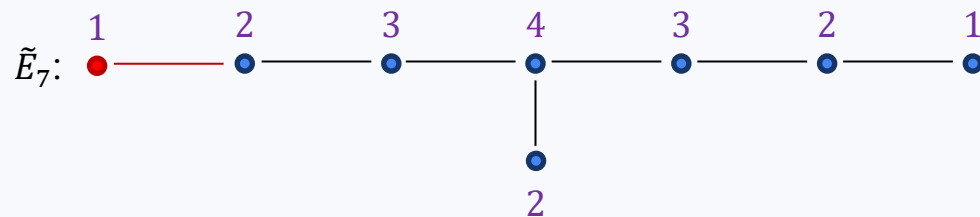
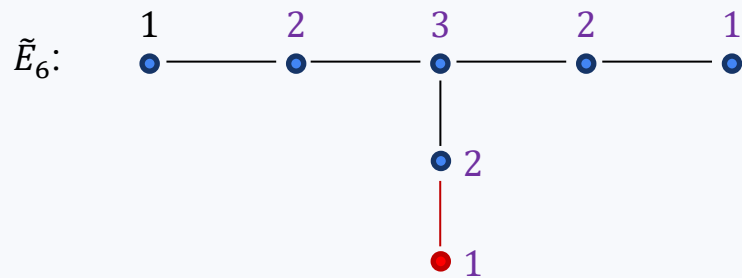
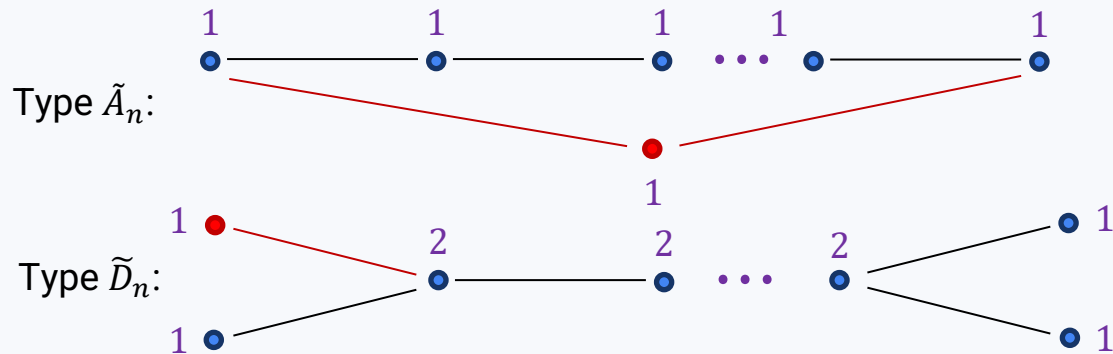


Indecomposables = positive roots



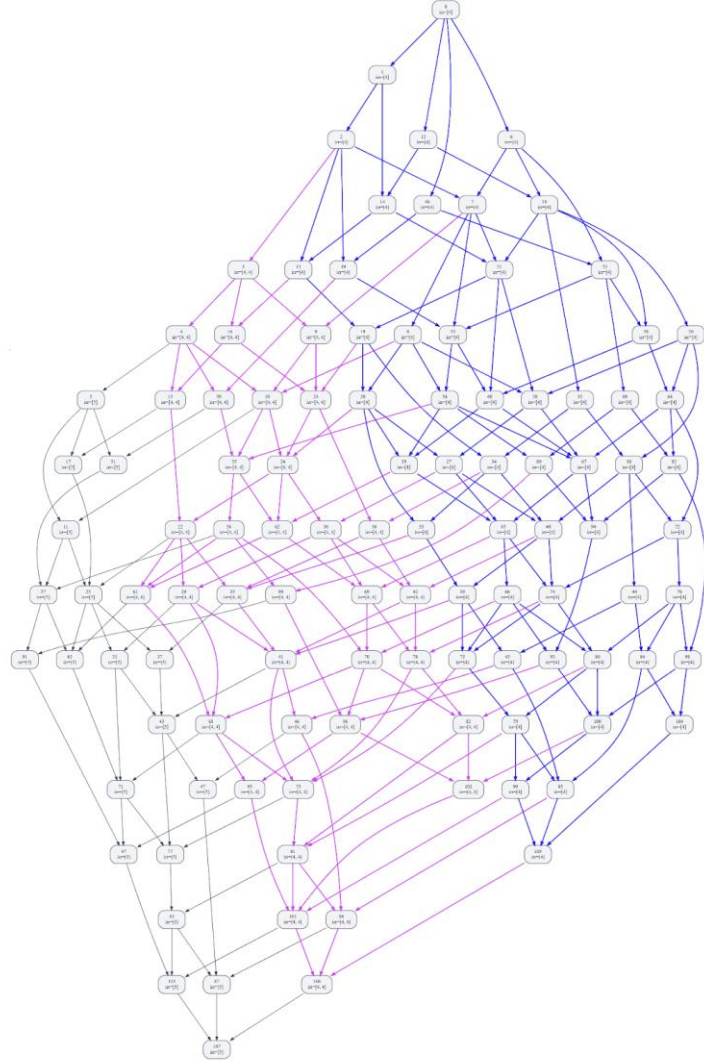
Tame quivers

Indecomposables =
real roots + imaginary roots



Degeneration graph

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$



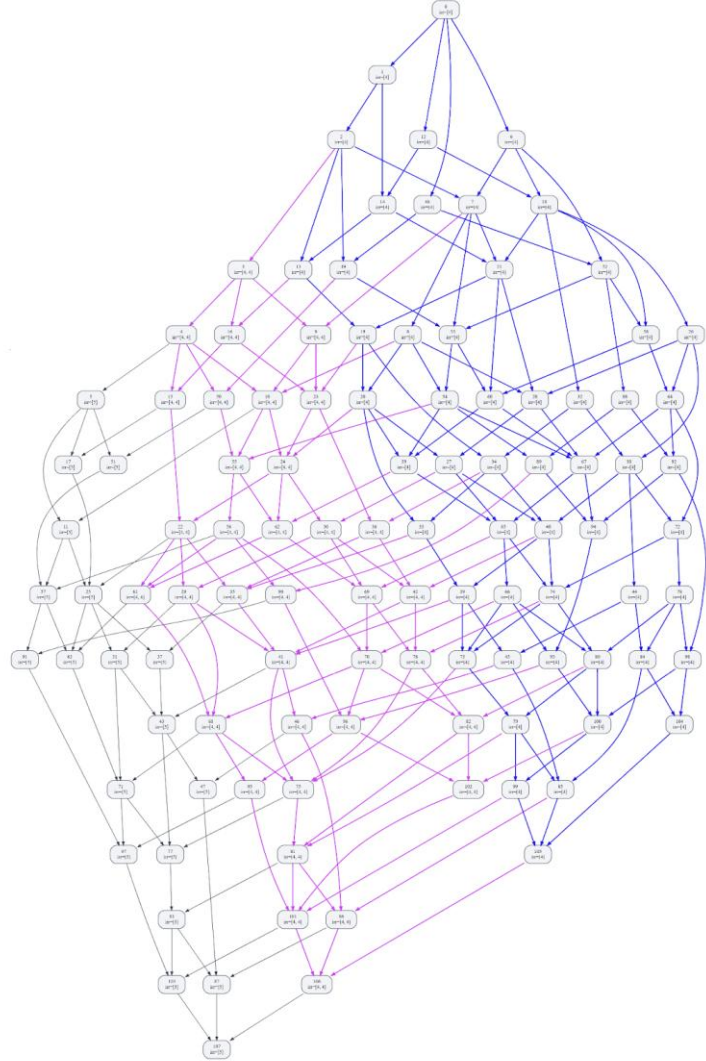
Degeneration graph

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

GL_d

1 t^{-1} t^{-1} t^{-1}



Degeneration graph

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

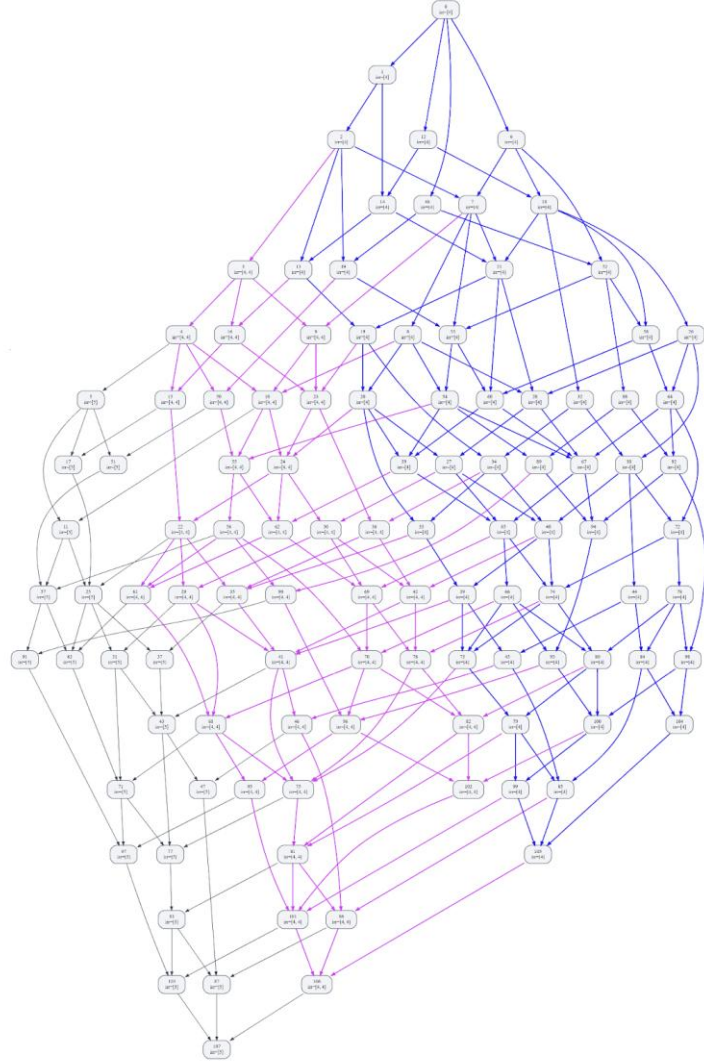
$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

GL_d

1 t^{-1} t^{-1} t^{-1}

$$(t^{-1})^{-1} \cdot 1 \cdot 1 = t$$

$$G(t): \mathbb{C} \xrightarrow{t} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$



Degeneration graph

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

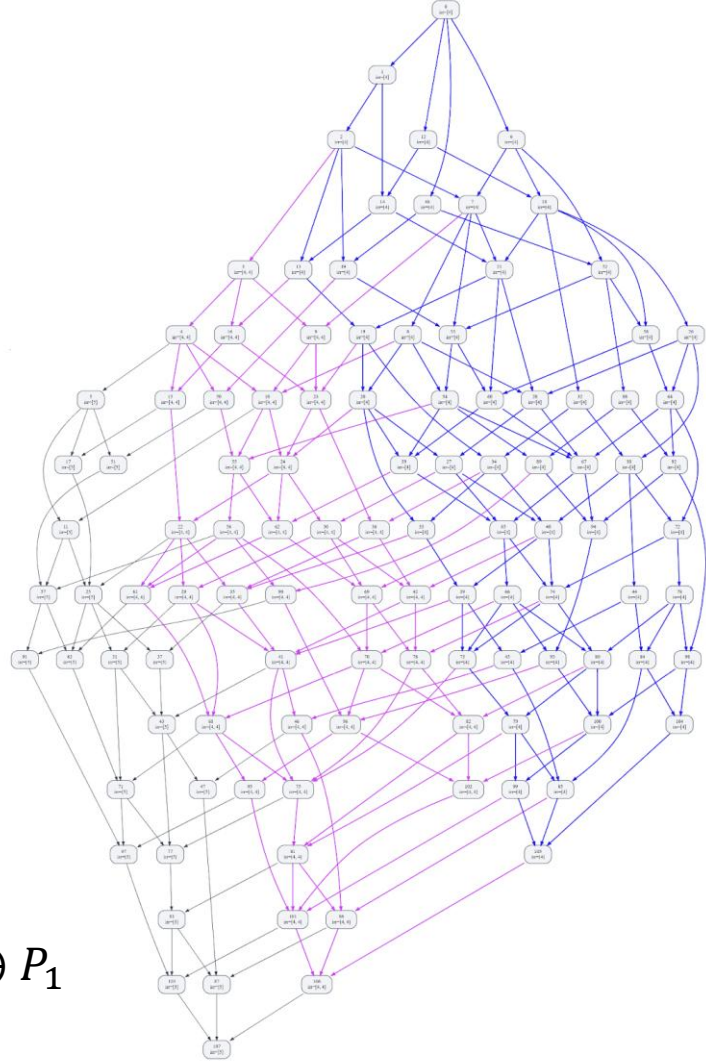
GL_d $\begin{matrix} \text{blue dashed arrows} \\ 1 \quad t^{-1} \quad t^{-1} \quad t^{-1} \end{matrix}$

$$(t^{-1})^{-1} \cdot 1 \cdot 1 = t$$

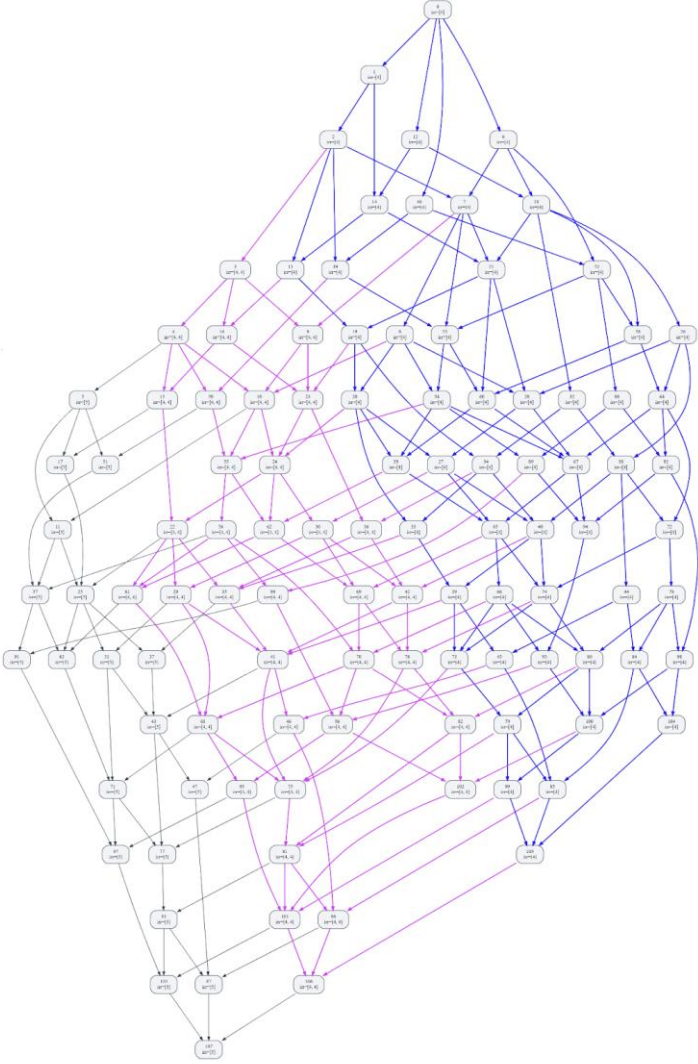
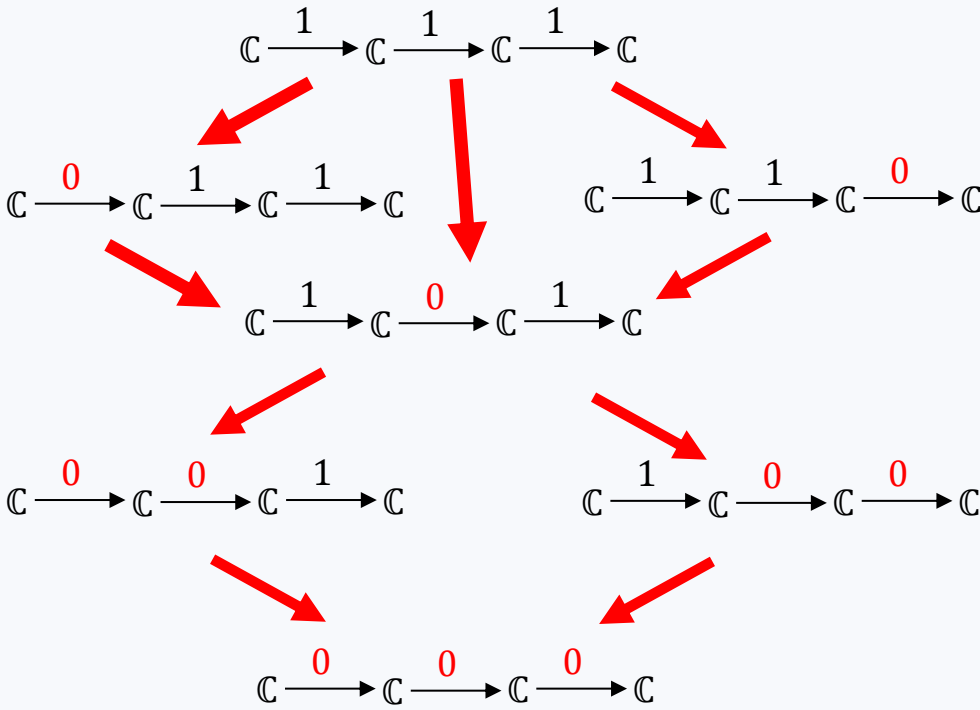
$$G(t): \mathbb{C} \xrightarrow{t} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$G(0): \mathbb{C} \xrightarrow{0} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$\overline{GL_d \cdot P_0} \cong S_0 \oplus P_1$$



Degeneration graph



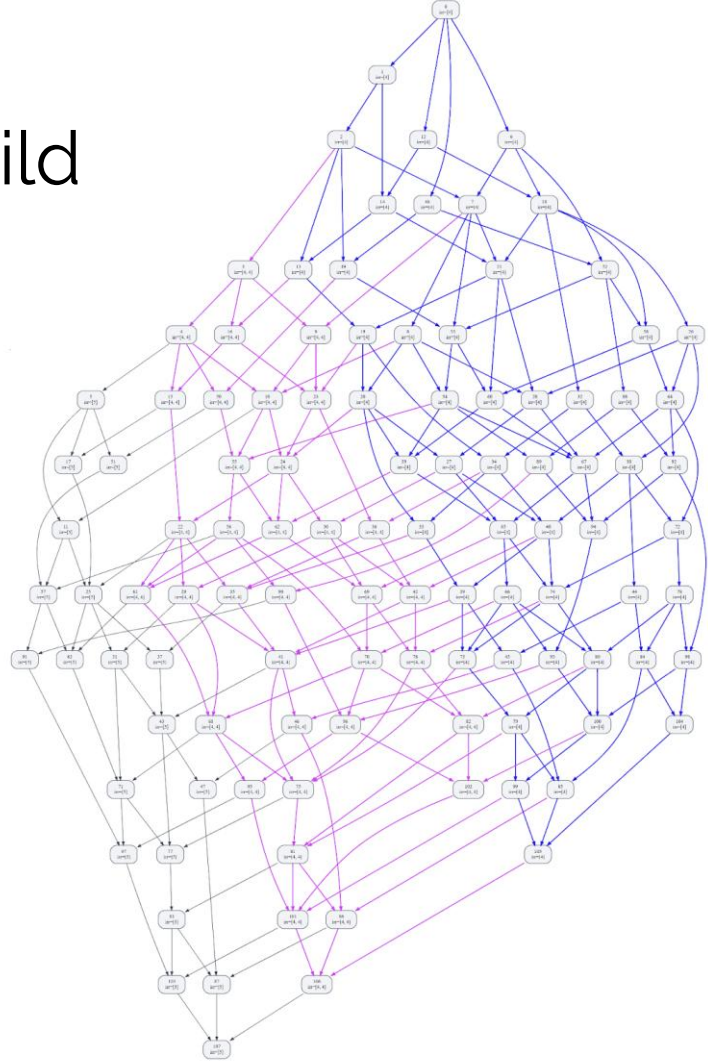
Degeneration graph – how to build

The Bongartz criterion

M degenerates to N if and only if, for every X ,

$$\dim \operatorname{Hom}_Q(M, X) \leq \dim \operatorname{Hom}_Q(N, X)$$

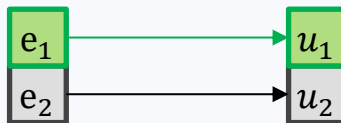
We can only check if for indecomposable X .



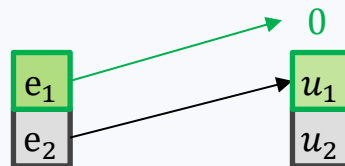
Subrepresentations

$N_i \subseteq M_i$, if $M_a(N_i) \subseteq N_j$
for all $a: i \rightarrow j$

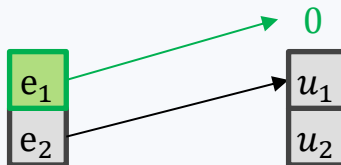
Subrepresentation



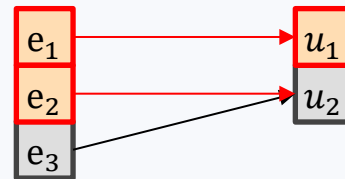
Subrepresentation



Subrepresentation

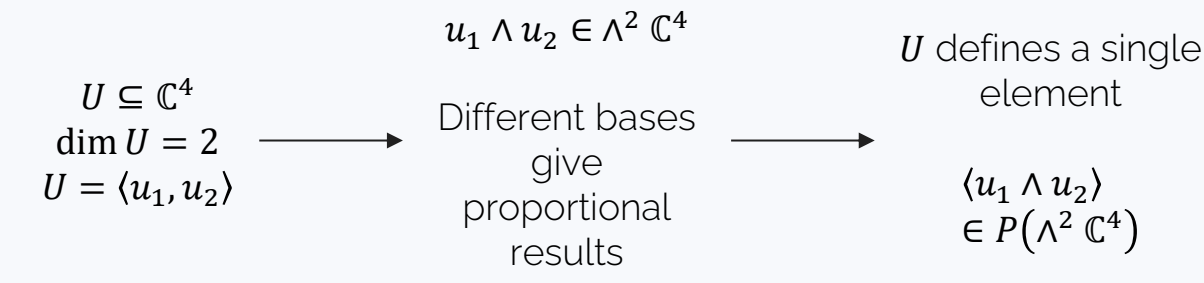


Not a
subrepresentation



Grassmannian

$\text{Gr}_d(\mathbb{K}^n)$ is the set of all d -dimensional subspaces of $\mathbb{K}^n$. It is a projective variety.



$\Lambda^2 \mathbb{C}^4$ has basis $e_i \wedge e_j, i \neq j$

Coordinates: $p_{12}, p_{13}, p_{14}, p_{23}, p_{24}, p_{34}$

$\text{Gr}_2(\mathbb{C}^4)$ is defined in $P(\Lambda^2 \mathbb{C}^4) \cong \mathbb{P}^6$ by a single equation:

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$$

Quiver Grassmannian

$\text{Gr}_d(M)$ is the set of all subrepresentations of M with dimension vector d .

Example.

$$M: \quad \mathbb{C}^5 \xrightarrow{I} \mathbb{C}^5 \xrightarrow{I} \mathbb{C}^5 \xrightarrow{I} \mathbb{C}^5$$

$$d = (1, 2, 3, 4)$$

Then $\text{Gr}_d(M)$ is the flag variety

$$\{U_1 \subseteq U_2 \subseteq U_3 \subseteq U_4 \subseteq \mathbb{C}^5 \mid \dim U_k = k\}$$

Quiver Grassmannian

$\text{Gr}_d(M)$ is the set of all d -dimensional subrepresentations of M .

$$\text{Gr}_d(M) \hookrightarrow \prod_i \text{Gr}_{d_i}(M_i)$$

Quiver Plücker relations. Let $m_{v,j,i}$ be the matrix coefficients of the linear maps $M_v : M_p \rightarrow M_q$. For a subset $I \cup \{i\}$ of $\mathcal{B}_p$, we define $\epsilon(i, I) = \#\{i' \in I \mid i' \leq i\}$. In this text, we show that the image of $\text{Gr}_{\underline{e}}(M)$ in $\prod \text{Gr}(e_p, M_p)$ is carved out by the *quiver Plücker relations*

$$E(v, I, J) = \sum_{i \in \mathcal{B}_p - I, j \in J} (-1)^{\epsilon(i, I) + \epsilon(j, J)} m_{v,j,i} \Delta_{I \cup \{i\}} \Delta_{J - \{j\}} = 0$$

Paths!

where $v : p \rightarrow q$ varies through the arrows of Q , I varies through the $(e_p - 1)$ -subsets of $\mathcal{B}_p$ and J varies through the $(e_q + 1)$ -subsets of $\mathcal{B}_q$.

What do we know about quiver Grassmannians?

Fact 1. Every projective variety is a quiver Grassmannian.

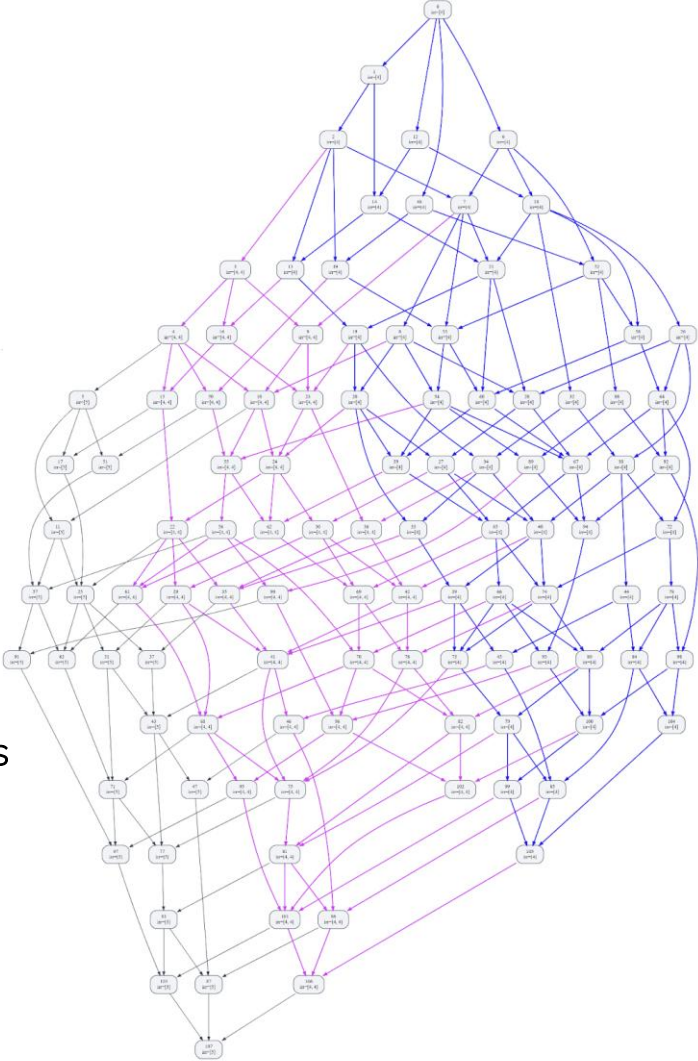
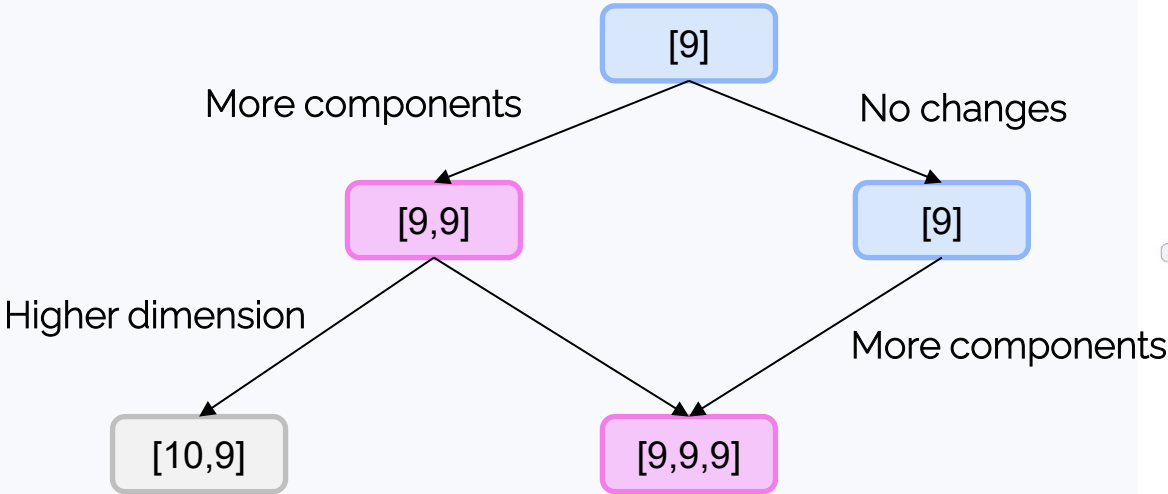
Fact 2. If Q is equioriented A_n , then $\mathrm{Gr}_{\dim(P^*)}(P^* \oplus I^*)$ is the PBW degenerate flag variety

Ordinary flag variety:

$$\mathcal{F}_{n+1} := \overline{G \cdot [v_{n+1}]} \subseteq \mathbb{P}(V_{n+1})$$

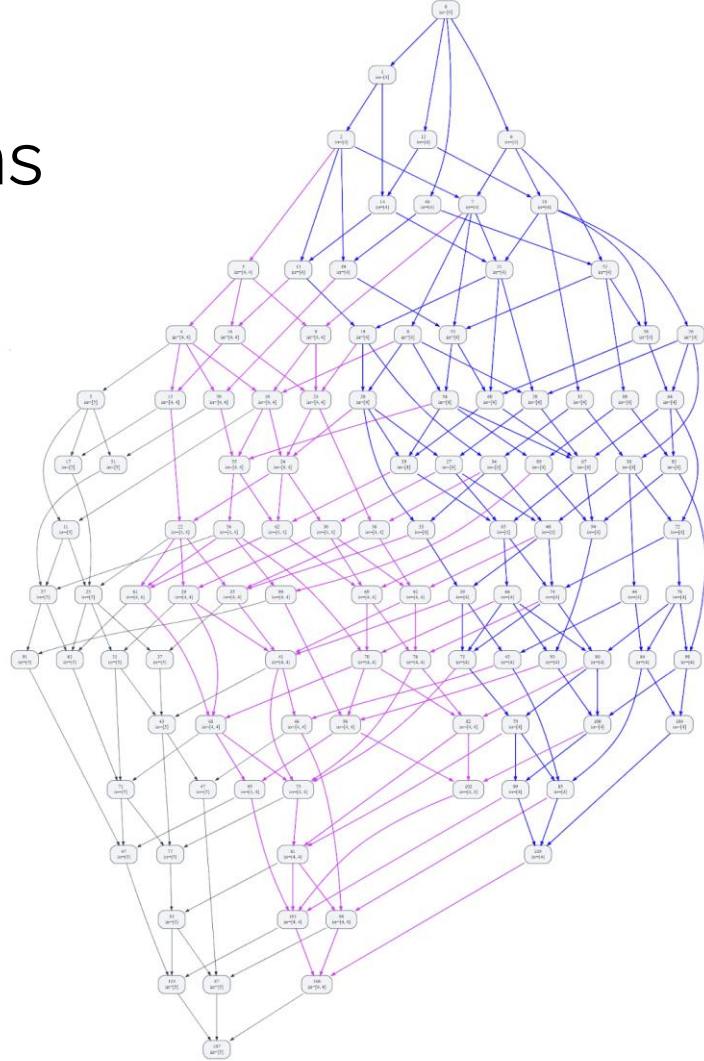
Now replace in the PBW theorem $U(\mathfrak{n}^-)$ by $S(\mathfrak{n}^-)$.

Degeneration and Grassmannians



Degeneration and Grassmannians

- M is **super cool** if $\text{Gr}_d(M)$ is irreducible of minimal dimension
- M is **cool** if $\text{Gr}_d(M)$ is of minimal dimension (not necessarily irreducible)





Projectives: all paths
from a given vertex

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_1: 0 \xrightarrow{0} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_2: 0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_3: 0 \xrightarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{C}$$



Projectives: all paths
from a given vertex

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_1: 0 \xrightarrow{0} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_2: 0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$P_3: 0 \xrightarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{C}$$

Injectives: all paths
to a given vertex

$$I_0: \mathbb{C} \xrightarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} 0$$

$$I_1: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{0} 0 \xrightarrow{0} 0$$

$$I_2: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{0} 0$$

$$I_3: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$



Projectives: all paths
from a given vertex

Injectives: all paths
to a given vertex

$$P_0: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$I_0: \mathbb{C} \xrightarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} 0$$

$$P_1: 0 \xrightarrow{0} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

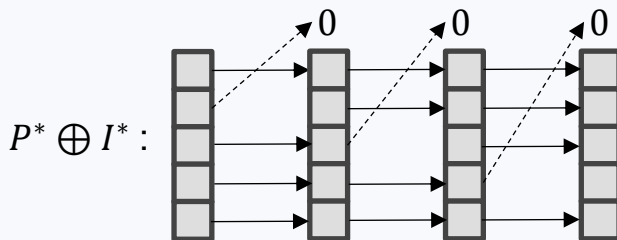
$$I_1: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{0} 0 \xrightarrow{0} 0$$

$$P_2: 0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

$$I_2: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{0} 0$$

$$P_3: 0 \xrightarrow{0} 0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{C}$$

$$I_3: \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C} \xrightarrow{1} \mathbb{C}$$

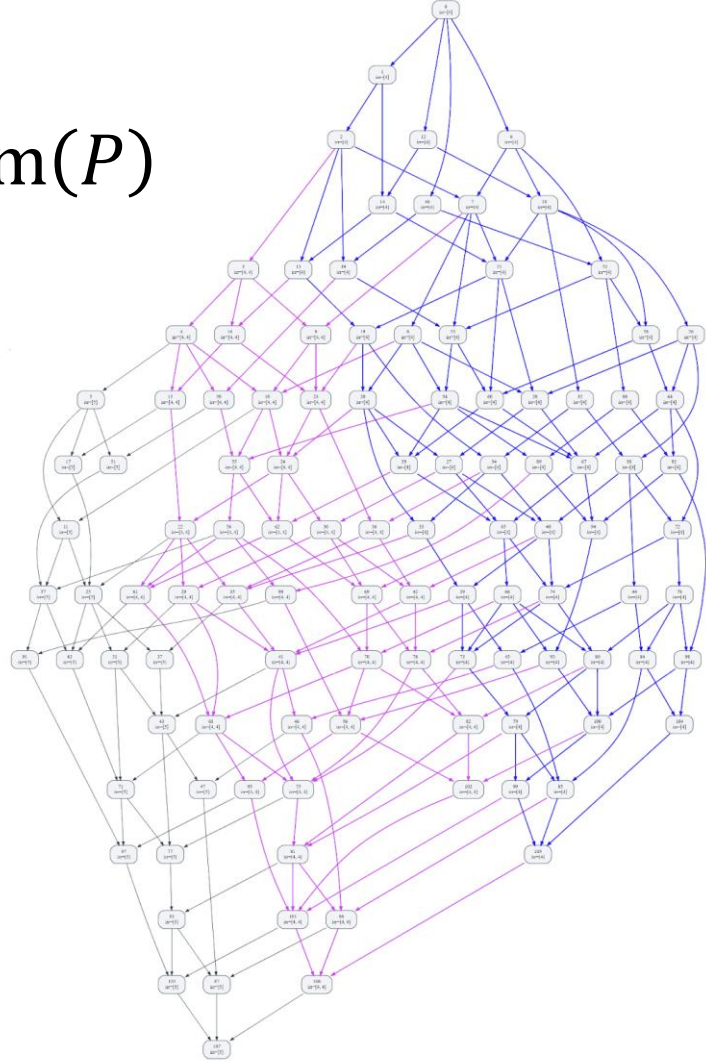


$$Q = A_n, \dim = \dim(P \oplus I), d = \dim(P)$$

- M is **super cool** if $\text{Gr}_d(M)$ is irreducible of minimal dimension
- M is **cool** if $\text{Gr}_d(M)$ is of minimal dimension (not necessarily irreducible)

Then:

1. If M degenerates to $P \oplus I$, then M is **super cool**.
2. All **super cool** representations degenerate to a single representation M^1 . This representation might be $P \oplus I$, but not necessary.
3. If furthermore, Q is *equi-directional*, then all **cool** M also degenerate to a single representation M^2 .



Here come coding agents

Agents: GPT, Codex, Claude Code

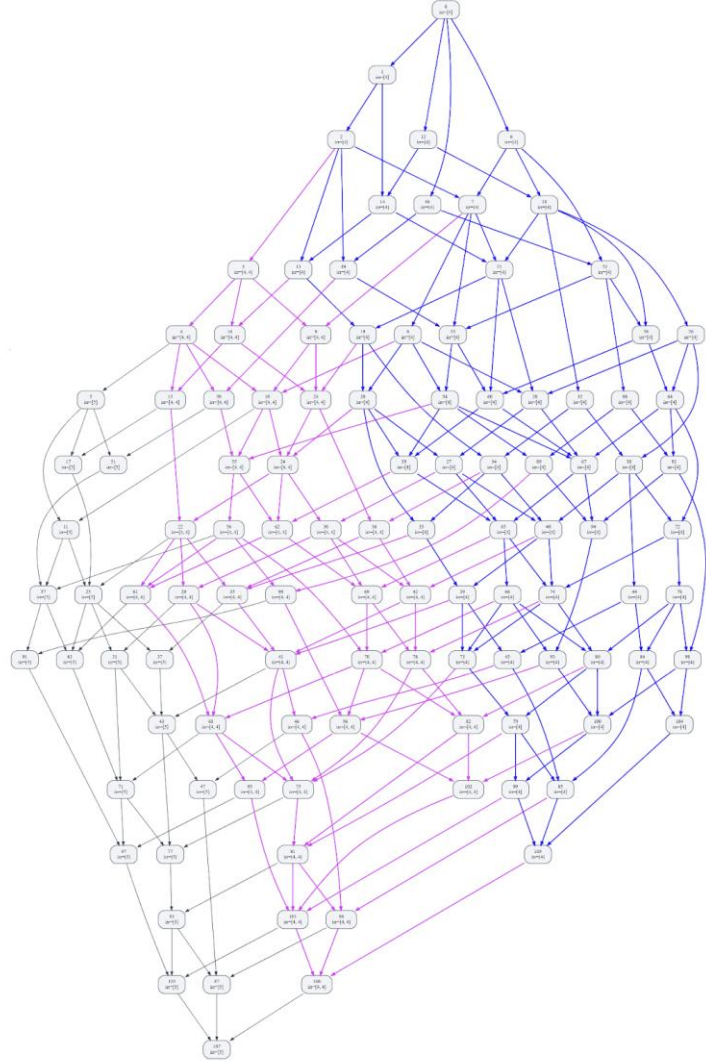
Tools: Python, Macaulay 2, a bit of Sage

Hardware: 128xCPU@Nebius

We've built a library for working with quivers

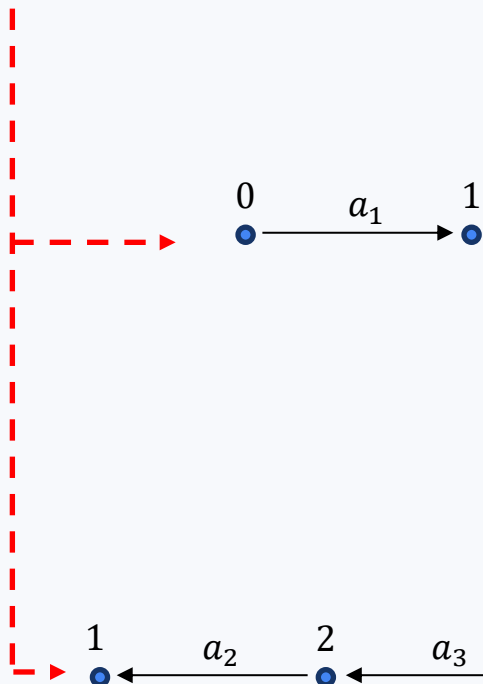
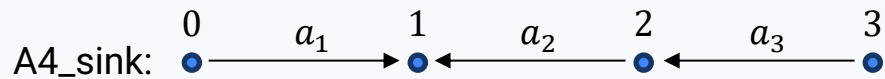
<https://github.com/st-fedotov/quiver/tree/main>

1. $M^1 = P \oplus I$ iff $\dim P$ and $\dim I$ have no zeros
2. For A_n , M^2 is always unique. For D_n , there might be 1 or 3.
3. And more!



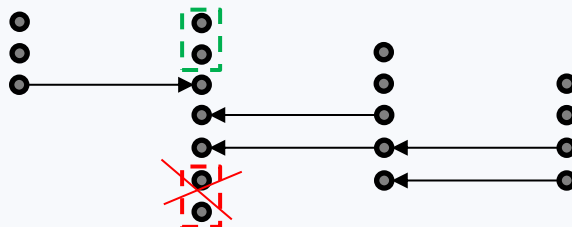
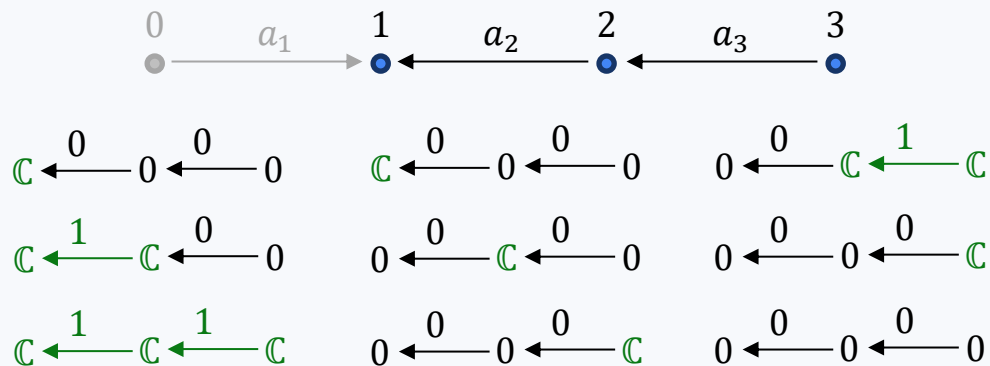
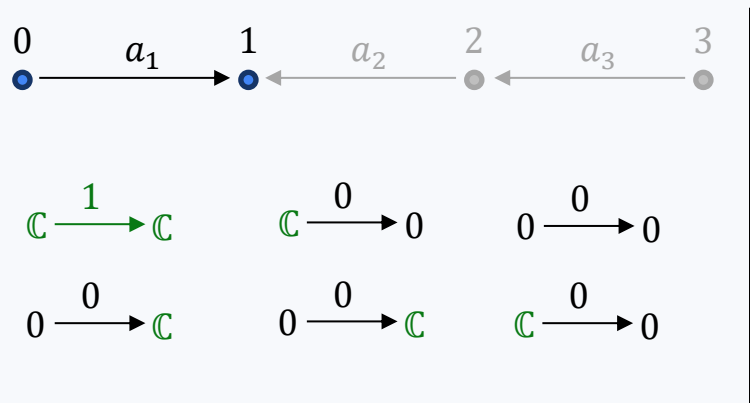
Lessons learnt

1. On the design stage – discussion first, then coding
2. Don't trust your coding agent in dealing with advanced maths
3. When debugging long pipelines, don't let your coding agent fixate on its terminal parts
4. I wish there were **skills** back then



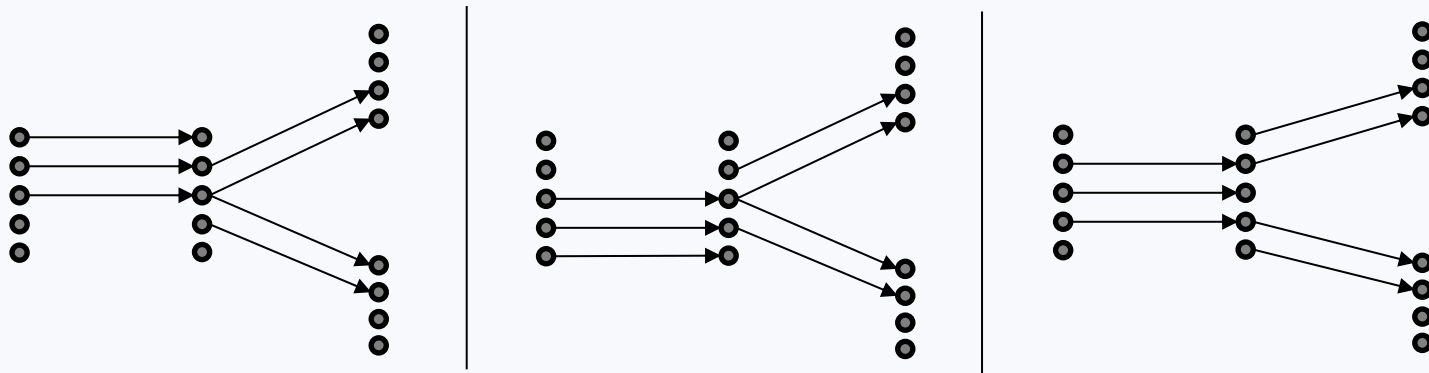
$$\begin{array}{c}
 \mathbf{P} \quad \oplus \quad \mathbf{S} \quad \oplus \quad \mathbf{I/S} \\
 \begin{array}{ccc}
 \mathbb{C} \xrightarrow{1} \mathbb{C} & \mathbb{C} \xrightarrow{0} 0 & \boxed{\begin{array}{c} \mathbb{C} \xrightarrow{0} 0 \\ \mathbb{C} \xrightarrow{1} \mathbb{C} \end{array}} \\
 0 \xrightarrow{0} \mathbb{C} & 0 \xrightarrow{0} \mathbb{C} & = \boxed{\begin{array}{c} 0 \xrightarrow{0} 0 \\ \mathbb{C} \xrightarrow{0} 0 \end{array}}
 \end{array}
 \end{array}$$

$$\begin{array}{c}
 \mathbf{P} \quad \oplus \quad \mathbf{S} \quad \oplus \quad \mathbf{I/S} \\
 \begin{array}{ccc}
 \mathbb{C} \xleftarrow{0} 0 \xleftarrow{0} 0 & \mathbb{C} \xleftarrow{0} 0 \xleftarrow{0} 0 & \boxed{\begin{array}{c} \mathbb{C} \xleftarrow{1} \mathbb{C} \xleftarrow{1} \mathbb{C} \\ 0 \xleftarrow{0} \mathbb{C} \xleftarrow{1} \mathbb{C} \\ 0 \xleftarrow{0} 0 \xleftarrow{0} \mathbb{C} \end{array}} \\
 \mathbb{C} \xleftarrow{1} \mathbb{C} \xleftarrow{0} 0 & 0 \xleftarrow{0} \mathbb{C} \xleftarrow{0} 0 & \\
 \mathbb{C} \xleftarrow{1} \mathbb{C} \xleftarrow{1} \mathbb{C} & 0 \xleftarrow{0} 0 \xleftarrow{0} \mathbb{C} &
 \end{array}
 \end{array}$$



Delete excessive simples
We should only have 2 at the seam

The three M^2 for D_4



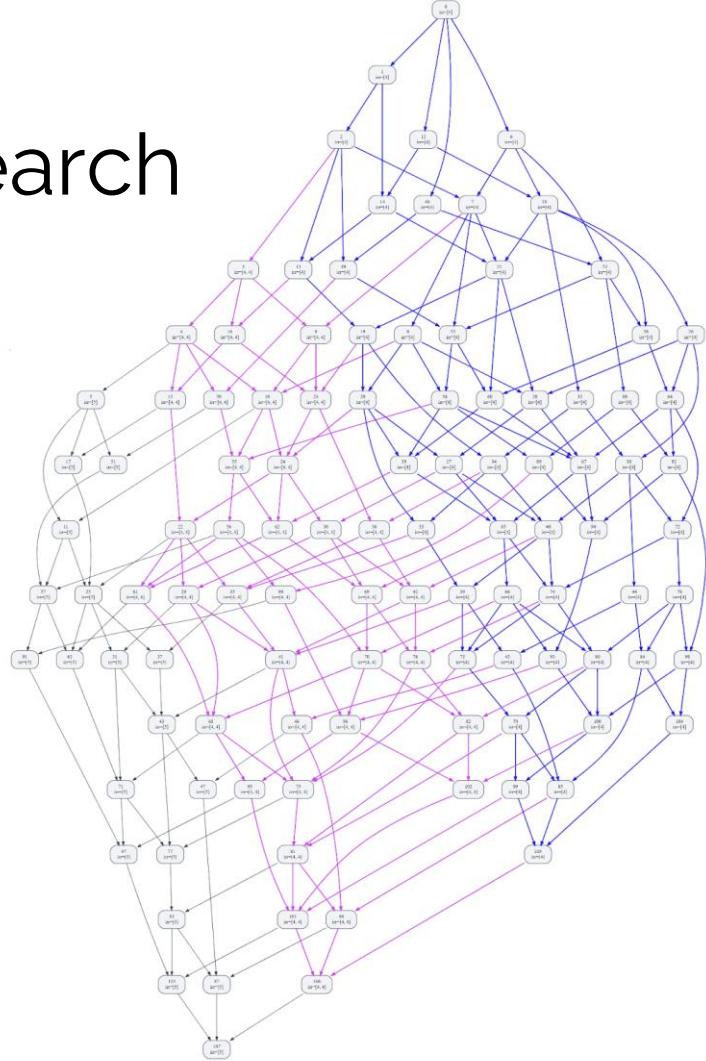
Nothing like its A_n counterparts!

What could possibly connect them?!

LLM-powered hypothesis search

Claude:

- For A_n quivers, if M consists entirely of single-vertex intervals $[i, i]$, then $Gr_d(M)$ is irreducible.
- For A_n zigzag quivers, modules with intervals of alternating parities (even-length and odd-length intervals alternating) have higher irreducibility rate than other orientations.



LLM-powered hypothesis search

Codex:

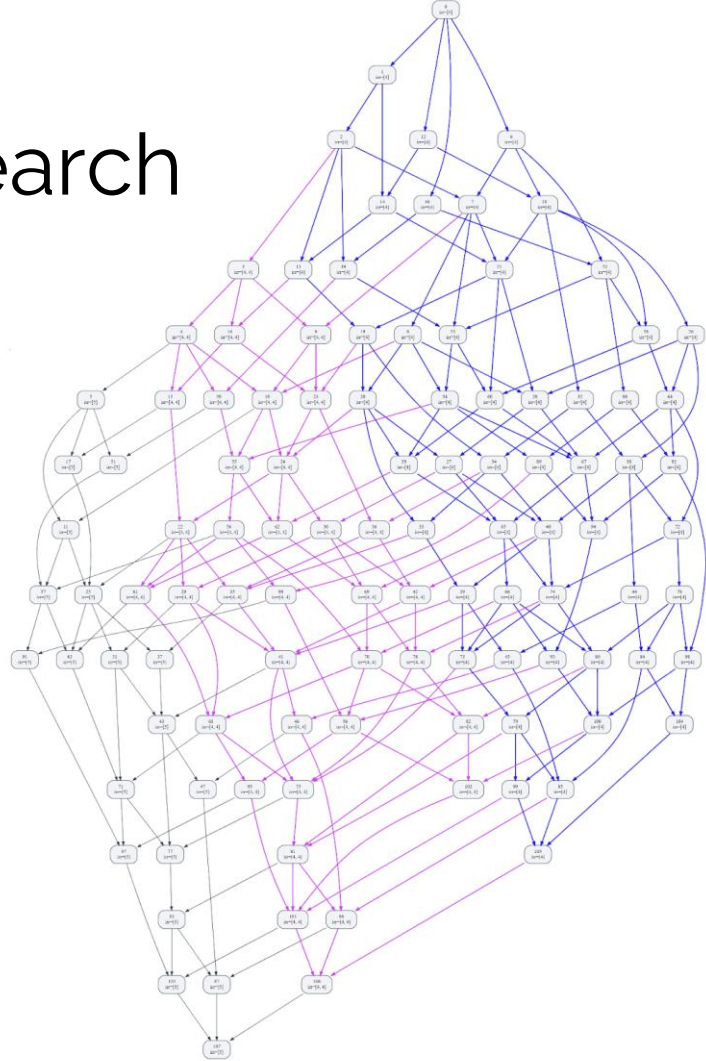
For A_n quivers, let M be a module with intermediate-dimension Grassmannian ($\min_dim < \dim Gr_d(M) < \max_dim$). Define the transitivity defect $T(M)$ as the number of triples (I, J, K) of intervals in M where I overlaps J , J overlaps K , but I does not overlap K .

If $T(M) = 0$, then $Gr_d(M)$ is irreducible.

Formal Definition:

Let $M = \sum I_i$ where $I_i = [a_i, b_i]$ are interval modules. Two intervals overlap if $\max(a_i, a_j) \leq \min(b_i, b_j)$. The transitivity defect counts violations of transitivity in the overlap relation:

$$T(M) = |\{(i, j, k) : I_i \text{ overlaps } I_j \text{ and } I_j \text{ overlaps } I_k \text{ but } I_i \text{ does not overlap } I_k\}|$$



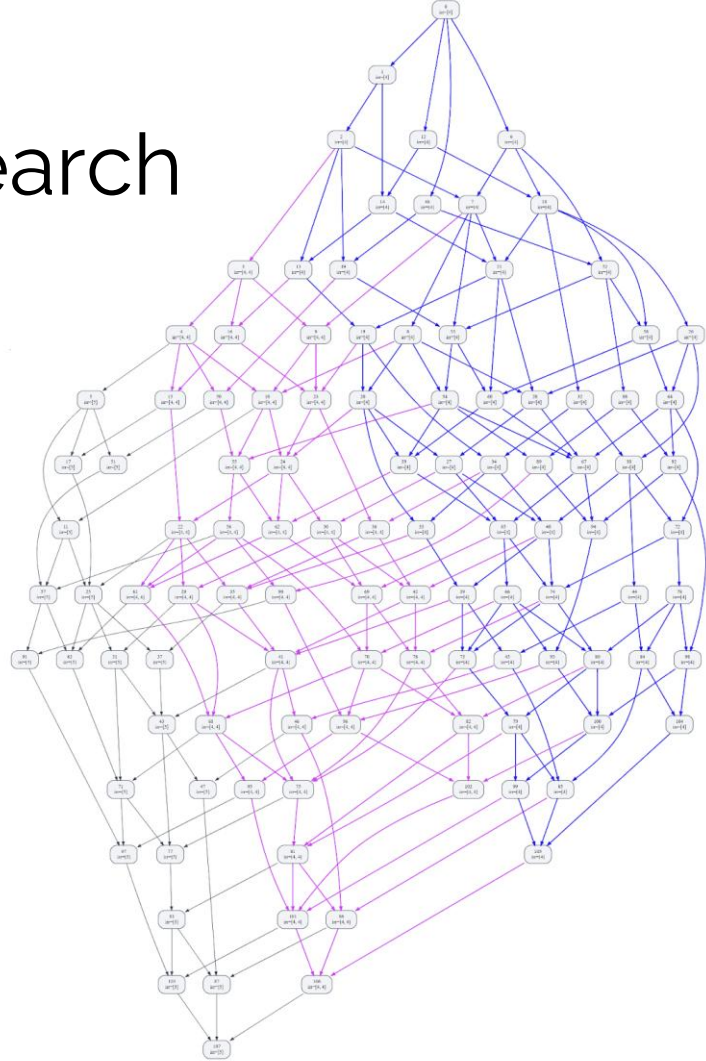
LLM-powered hypothesis search

Claude again:

For A_n quivers with intermediate modules M , the number of irreducible components of $Gr_d(M)$ exhibits a sharp dichotomy based on the transitivity defect:

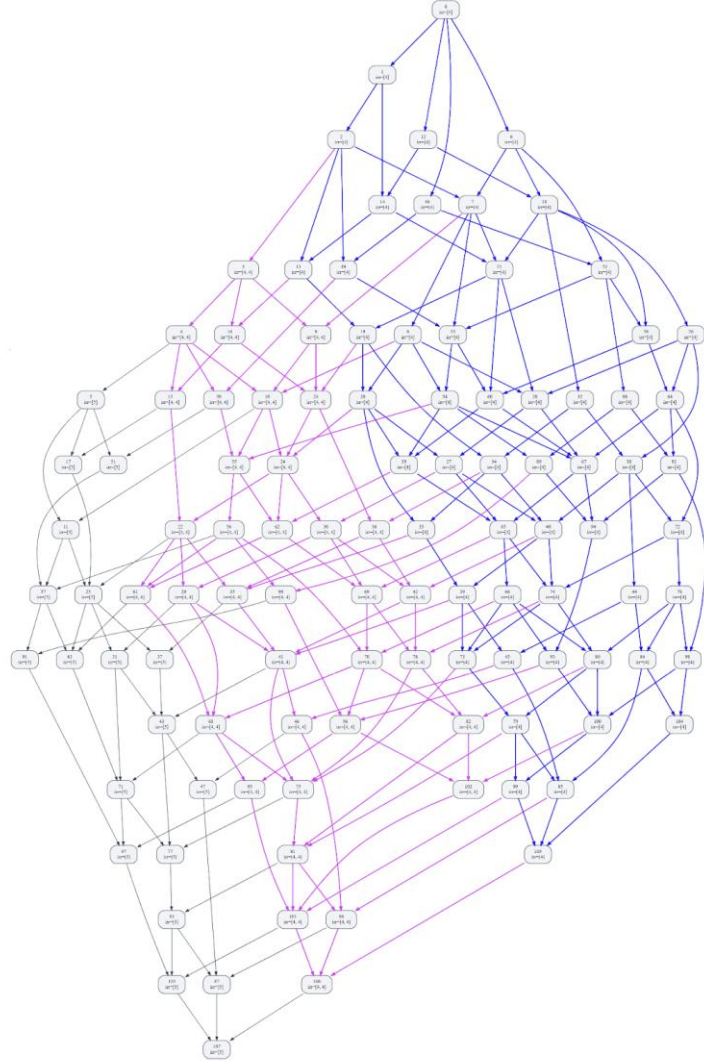
- If $T(M) = 0$: $Gr_d(M)$ is always irreducible (1 component)
- If $T(M) > 0$: $Gr_d(M)$ has on average 6.42 components (96% reducible)

The transitivity defect acts as a complexity measure for geometric decomposition, with a phase transition at $T(M) = 0$.



Need more data!

- Computed $\text{Hom}_Q(M, X)$ from Bongartz theorem across all indecomposables X
- Noticed a pattern
- Ran many experiments with Claude Code, refined and validated hypotheses
- Profit!



The Hom criterion

For both A_n and D_n , unless $P \oplus I$ is deficient:

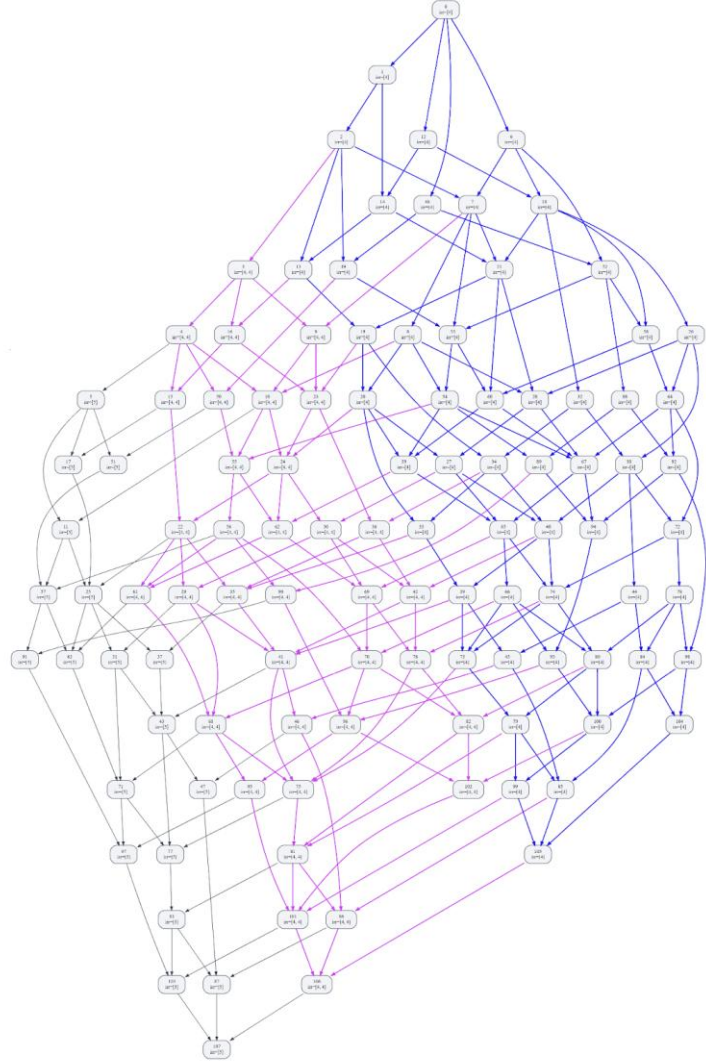
1. M is **super cool** iff, for every non-injective X ,

$$\dim \operatorname{Hom}_Q(M, X) \leq \dim \operatorname{Hom}_Q(P, X)$$

(obvious from the Bongartz theorem, since for a non-injective indecomposable X : $\dim \operatorname{Hom}_Q(I, X) = 0$)

2. M is **cool** iff, for every non-injective X ,

$$\dim \operatorname{Hom}_Q(M, X) \leq \dim \operatorname{Hom}_Q(P, X) + 1$$



Some obstacles we've faced

1. Combinatorial explosion: can't really compute things for A_5 or for $P \oplus I$ with multiplicities > 2 .
2. Rare events: I've only observed 3 cases of violation of the Hilbert function hypothesis (feeling like a high energy physicist).
3. LLMs as research agents face difficulties when the right conjecture features **input restrictions** ($P \oplus I$ non-deficient) and/or **data-dependent formulas** ($\dim \operatorname{Hom}_Q(P, X) + 1$ threshold).

Some further plans

1. Describe M^2 in corner cases
2. Move on to other dimension vectors **On it!**
3. Move on to tame quivers